

Graphical abstract



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Lysosomal dysfunction is associated with neurodegeneration. Balak et al. examine the cell-specific effects of lysosomal dysfunction in an MPS-IIIa (*Sgsh*^{-/-}) model and find that microglia are disproportionately affected.

Mechanistically, the MITF/TFE transcription factors cooperate with AP-1/ATF, C/EBP, and PU.1 to drive a lysosome-dependent program within microglia that is shared across neurodegeneration models and Alzheimer's disease.

Highlights

- Microglia are the most impacted cells in an MPS-IIIa lysosomal storage disease model
- MITF/TFE cooperate with AP-1/ATF, C/EBP, and PU.1 to activate microglial enhancers
- ASOs targeting *Mitf* + *Tfe3* partially reverse the MPS-IIIa microglial program *in vivo*
- The MPS-IIIa microglial program is conserved in neurodegeneration models and human AD

Article

Lysosomal dysfunction drives a transcriptional and epigenetic signature found in disease-associated microglia in neurodegenerative diseases

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SUMMARY

Lysosomal dysfunction is causally linked to neurodegeneration in many lysosomal storage disorders and is associated with various age-related neurodegenerative diseases. Here, we investigated the question of underlying mechanisms using a mouse model of mucopolysaccharidosis type IIIA caused by deficiency of the lysosomal hydrolase SGSH. Systematic imaging and transcriptomic and epigenetic studies revealed microglia to be the most profoundly impacted cell type in brains of *Sgsh*-deficient mice. Further investigation identified dominant and context-dependent roles of members of the MITF/TFE family as major drivers of microglia-specific epigenetic and transcriptional changes resulting from lysosomal stress that are dependent on collaborative interactions with AP-1/ATF, C/EBP, and PU.1/ETS transcription factors. Features of the transcriptomic and epigenetic alterations observed in murine *Sgsh* deficiency were also observed in microglia derived from mouse models of age-related neurodegeneration and in human Alzheimer's disease patients. These findings reveal common and disease-specific transcriptional mechanisms associated with disease-associated microglia phenotypes.

INTRODUCTION

Pathogenic variants causing lysosomal storage diseases (LSDs) are a leading cause of childhood neurodegeneration and dementia,^{1,2} indicating that lysosomal defects alone are sufficient to drive pathological processes that result in neuronal dysfunction and death. Beyond macromolecule degradation, lysosomes

act as metabolic and inflammatory signaling hubs regulating autophagy, energy homeostasis, inflammation, and phagocytosis in a cell type-specific manner.^{3–5} Consistent with these functions, autophagosomal and endolysosomal dysfunction is implicated in age-related neurodegenerative diseases^{6–10} including Alzheimer's disease (AD),^{11–14} Parkinson's disease (PD),^{15,16} frontotemporal dementia (FTD), and amyotrophic



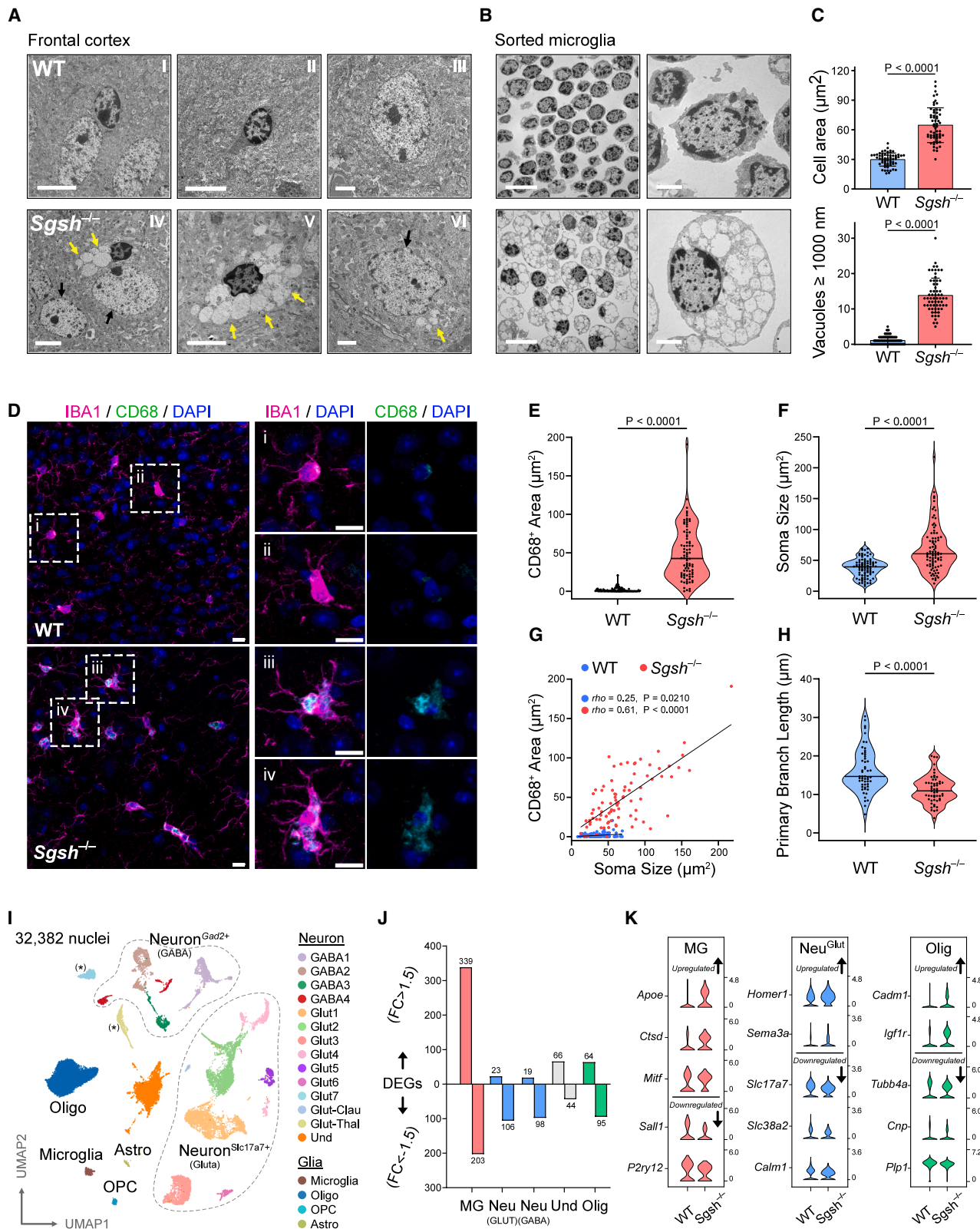


Figure 1. *Sgsh* deficiency preferentially affects lysosomal morphology and gene expression in microglia

(A) EM images (cortex) from 8-month-old WT and *Sgsh*^{-/-} mice (*n* = 4 mice/genotype). Yellow arrows, expanded vacuoles; black arrows, neuronal nuclei.
(B) EM images (sorted microglia) from 6-month-old WT and *Sgsh*^{-/-} brains (*n* = 3 mice/genotype).

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lateral sclerosis (ALS),^{17–22} and risk variants for these disorders frequently affect genes involved in endolysosomal biology.^{23–33}

Lysosomal dysfunction can also result from local environmental perturbations, including uptake of degradation-resistant protein aggregates or toxic particles that disrupt lysosomal integrity.^{34,35}

In LSDs, all cells carry the pathogenic variant, yet the cell type-specific consequences of losing an essential lysosomal function remain poorly understood. Although some effects are expected to be cell autonomous in neurons, neuronal dysfunction/death may also result from effects in other cell types, such as microglia, which require robust lysosomal capacity to process autophagic and phagocytic cargos and maintain neural homeostasis. While microglia play protective roles through clearance of debris and pathological protein aggregates, disease-associated lysosomal dysfunction can shift these cells toward maladaptive inflammatory/neurotoxic programs that contribute to neuropathology.^{36–38} Despite these documented roles, the transcriptional mechanisms by which loss of an essential lysosomal enzyme reshapes gene expression programs across brain cell types have not been established for any LSD. Defining these mechanisms may provide insight into LSD pathogenesis and shared responses to lysosomal stress in age-related neurodegenerative diseases.

While most age-related neurodegenerative diseases are genetically complex, monogenic LSDs provide a tractable framework for studying the consequences of intrinsic lysosomal dysfunction across brain cell types. Mucopolysaccharidosis type IIIA (MPS-IIIa; Sanfilippo syndrome type A) is a severe neurodegenerative LSD caused by biallelic loss-of-function mutations in *SGSH*, a lysosomal sulfamidase required for heparan sulfate (HS) degradation.^{39,40} Affected children develop progressive behavioral, motor, and cognitive impairment/dementia, and typically succumb to disease by late adolescence.^{41–49} *Sgsh*^{−/−} mice recapitulate key disease features, including reduced lifespan and progressive neurocognitive and behavioral impairments.^{50,51} In humans and mice, *SGSH* deficiency causes lysosomal accumulation of HS, a major component of the brain extracellular matrix (ECM) which microglia actively remodel.^{50,52–55} Shared neuropathology includes microgliosis, astrogliosis, and secondary macromolecule/lipid accumulation (e.g., GM2/GM3 gangliosides, cholesterol),^{50,56–61} with mouse models further showing pathogenic protein aggregates and synaptic abnormalities.^{59,60,62} Disease-modifying therapies remain elusive,⁶³ highlighting the need to better understand how *SGSH* deficiency affects different brain cells. Here, we sought to determine how loss of an essential lysosomal enzyme reshapes transcriptional and epigenetic programs across the major cell types of the brain, using the *Sgsh*^{−/−} (MPS-IIIa) mouse model.

We find that microglia are disproportionately affected and identify MITF and TFE3 transcription factors (TFs) as major drivers of the disease-associated microglial gene expression program—a regulatory network that is also embedded in microglial states observed in mouse models of age-related neurodegeneration and in human AD.

RESULTS

Sgsh deficiency preferentially affects lysosomal morphology and gene expression in microglia

To assess the consequences of *Sgsh* deficiency on brain ultrastructure, we analyzed frontal cortex of wild-type (WT) and *Sgsh*^{−/−} mice by transmission electron microscopy (EM). In *Sgsh*^{−/−} mice, we observed markedly enlarged (>1000 nm), electron-lucent vacuoles most prominently in cells bearing small, electron-dense nuclei; an ultrastructural pattern consistent with microglia and lysosomal expansion (Figures 1A IV–V, yellow arrows, and S1A). In contrast, cells exhibiting neuronal, astrocytic, or oligodendrocytic features^{64,65} were associated with smaller vacuolar structures (Figure 1A VI, yellow arrow) or no detectable vacuoles (black arrows). Secondary intraluminal material (e.g., lamellar-like structures) was observed in microglia and in the subset of neurons that exhibited smaller abnormal vacuoles (Figure S1B I–II). To verify highly vacuolated cells were microglia, CD11b⁺/CD45^{int}/CX3CR1⁺ cells were sorted from forebrains and analyzed by EM. Sorted microglia similarly exhibited small electron-dense nuclei, while *Sgsh*^{−/−} cells showed numerous abnormal vacuoles resembling those *in vivo* (Figure 1B), were approximately 2-fold larger than WT cells, and contained on average >10 enlarged vacuoles per cell (Figure 1C). *Sgsh*^{−/−} microglia also occasionally contained intracytoplasmic inclusions resembling dystrophic neurites,⁶⁴ suggesting recent phagocytosis and/or reduced digestion of damaged neuronal substructure (Figure S1B III–IV). By flow cytometry, *Sgsh*^{−/−} microglia exhibited increased CD45 and LysoTracker staining versus WT or *Sgsh*^{+/−} mice, consistent with a reactive phenotype and increased acidic organelle mass/number (Figures S1C and S1D).

Supporting these ultrastructural and flow cytometry results, immunofluorescence microscopy of cortex tissue examining the pan-microglial marker IBA1 and reactive microglia-associated phagolysosomal marker CD68 revealed markedly increased CD68 staining in IBA1⁺ *Sgsh*^{−/−} microglia (Figures 1D and 1E). Quantitative analysis indicated increased microglial soma size (Figure 1F) which positively correlated with CD68-positive area (Figure 1G), indicative of a lysosomal identity of the enlarged

(C) Quantification from (B): cell area (top) and vacuoles >1000 nm /cell.

(D) Immunofluorescence images in 6-month-old WT and *Sgsh*^{−/−} cortex. IBA1 (magenta), CD68 (green), DAPI (blue).

(E and F) Microglial CD68⁺ area and IBA1-based soma size (*n* = 2 mice/genotype).

(G) Correlation between soma size and CD68⁺ area. *p* values: Spearman's rank correlation.

(H) Microglia primary branch length by IBA1 staining, averaged per cell (*n* = 2 mice/genotype).

(I) UMAP of snRNA-seq from 8-month-old WT, *Sgsh*^{+/−}, and *Sgsh*^{−/−} forebrains (*n* = 2 mice/genotype). Circles, neuronal subclasses by neurotransmitter; asterisks, subcortical neurons.

(J) DEGs (*FC* > 1.5, *p*-adj < 0.05) in major brain cell types/clusters with >500 nuclei. Neu(GLUT), combined *Slc17a7*⁺ subclusters; Neu(GABA), combined *Gad2*⁺ subclusters; Und, unclear identity.

(K) Representative DEGs (*p*-adj < 0.05) in microglia (MG), oligodendrocytes (Olig), and glutamatergic neurons (Neu^{Glut}).

Scale bars: 5 μm (A, left/middle), 10 μm (B, left and D), and 2 μm (A, right and B, right). Dots represent individual cells (C and E–I). Lines, medians; *p* values, unpaired two-tailed *t* tests (C, E, F, and H).

See also Figure S1.

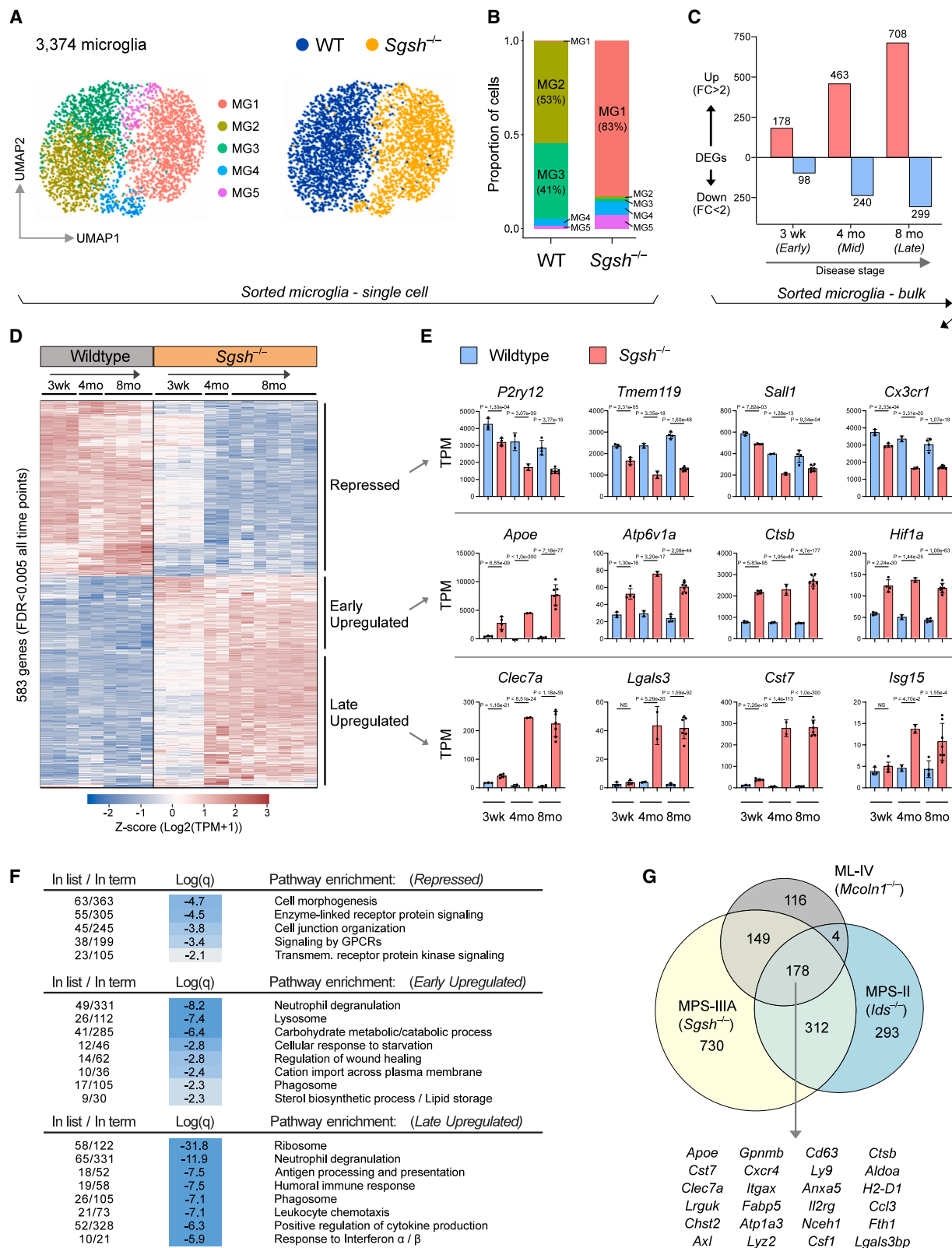


Figure 2. Time-dependent changes in microglial gene expression in *Sgsh*-deficient mice

(A) Uniform manifold approximation and projection (UMAP) of microglia population from CD45⁺ cells from 6-month-old WT and *Sgsh*^{-/-} forebrains, colored by subcluster and genotype (*n* = 2 mice/genotype).

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vacuoles. In parallel, primary branch lengths of microglial processes were reduced, consistent with microglia in a reactive state (Figure 1H).

To investigate the transcriptional consequences of *Sgsh* deficiency in the brain, we performed single-nucleus RNA sequencing (snRNA-seq) on forebrains from WT, *Sgsh*^{+/-} and *Sgsh*^{-/-} mice with late-stage disease (8 months). After quality control, ~32,300 nuclei were clustered then annotated by reference-based cell label transfer⁶⁶ (Figures 1I and S1E). For differential expression analysis, neuronal subpopulations were consolidated into inhibitory/GABAergic (*Gad2*⁺) and excitatory/glutamatergic (*Slc17a7*⁺) classes (Figure S1F), and clusters containing >500 nuclei were compared between disease (*Sgsh*^{-/-}) and unaffected (WT, *Sgsh*^{+/-}) samples using NEBULA.⁶⁷ Using an absolute fold change (FC) > 1.5 and adjusted *p* value (*p*.adj) < 0.05, microglia were the most affected cell type exhibiting >530 differentially expressed genes (DEGs), 339 of which were upregulated (Figure 1J). Neurons and oligodendrocytes exhibited ~3–5-fold fewer DEGs than microglia, a pattern consistent with findings in other age-related and lysosomal-associated neurodegenerative disorders.^{68,69} Upregulated microglial genes included those commonly induced in human/mouse neurodegenerative contexts (e.g., *Apoe*, *Cd9*, *Ctsd*, and *Mitf*), whereas downregulated genes included those associated with a homeostatic phenotype (*P2ry12* and *Sall1*) (Figure 1K). Oligodendrocyte DEGs included upregulated *Cadm1* and *Igf1r* and downregulated *Tubb4a*, *Cnp*, and *Plp1*, genes similarly dysregulated in other neurodegenerative and/or demyelination contexts.^{70,71} Together, these data reveal disproportionate morphological and transcriptional effects on microglia in the *Sgsh*-deficient brain.

Time-dependent changes in microglial gene expression in *Sgsh*-deficient mice

Although microglia exhibited pronounced global changes in whole-brain snRNA-seq, their relatively low recovery limited analysis of response heterogeneity. We therefore performed single-cell RNA sequencing (scRNA-seq) on sorted CD45⁺ cells from WT and *Sgsh*^{-/-} forebrains. Following quality control, 4,021 cells yielded 10 clusters (Figures S2A and S2B). Approximately 85% of cells were microglia, characterized by high *Tmem119* and *P2ry12* expression, with the remainder comprising minor populations of myeloid cells. Microglia further subclustered into 5 populations (MG1–MG5) which varied considerably in their genotype composition (Figures 2A and 2B). Notably, 94% of WT microglia localized to homeostatic MG2 and MG3 clusters marked by high *Tmem119*, *P2ry12*, and *Cx3cr1* expression (Figure S2C), compared with only ~4% of *Sgsh*^{-/-} microglia.

Conversely, 83% of *Sgsh*^{-/-} microglia, but only 1% of WT microglia, resided in MG1, a cluster that exhibited a disease-associated microglia (DAM)/microglia neurodegenerative (MGnD)-like signature,^{72,73} including downregulated homeostatic markers and upregulated *Apoe*, *Cst7*, *Cd9*, and *Ctsb* (Figure S2C). Subsequent differential expression analysis of microglia identified >1000 DEGs in *Sgsh*^{-/-} cells (FC > 1.5, *p*.adj < 0.05).

The predominance of MG1 in *Sgsh*^{-/-} microglia indicated a near-global state transition, contrasting other neurodegeneration models in which DAM/MGnD phenotypes represent a fraction of total microglia.^{72,73} We therefore utilized bulk RNA-seq on CD11b⁺/CD45^{int}/CX3CR1⁺ sorted microglia from WT and *Sgsh*^{-/-} brains (Figure S2D) at 3 weeks and 4 and 8 months of age to deeply capture gene expression changes across disease progression. *Sgsh*^{-/-} microglia exhibited a progressive increase in DEGs across timepoints, with a molecular phenotype already evident at 3 weeks (Figure 2C; Table S1). Upregulated genes (FC > 2, *p*.adj < 0.05) increased from 178 at 3 weeks to 708 at 8 months, while downregulated genes progressed from 98 to 299 over the same time interval. A heatmap of the 583 genes differentially expressed across all time points (*p*.adj < 0.005) is shown in Figure 2D. Approximately 40% of these exhibited progressive downregulation from 3 weeks to 4 months and remained repressed at 8 months, including *P2ry12*, *Tmem119*, and *Sall1* (Figure 2E, top). Pathway enrichment analysis of this group indicated modest enrichment for terms including cell morphogenesis, receptor signaling, and cell junction organization (Figure 2F, top).

Upregulated genes fell into two general classes. The first, defined by early upregulation at 3 weeks that remained elevated at 4 and 8 months, included *Apoe*, *Atp6v1a*, *Ctsb*, and *Hif1a* (Figure 2E, middle). Pathway enrichment of this group yielded predominantly lysosomal terms including neutrophil degranulation (a term enriched for lysosomal and vesicular trafficking genes), lysosome, phagosome, carbohydrate metabolism, and wound healing (Figure 2F, middle). The second class showed minimal upregulation at 3 weeks but robust upregulation at 4 and 8 months, and included *Clec7a*, *Lgals3*, *Cst7*, and *Isg15* (Figure 2E, bottom). This group showed further enrichment of lysosomal pathways but also inflammatory terms, including cytokine production and interferon response (Figure 2F, bottom). Consistent with this pattern, the number of upregulated genes linked to these inflammatory terms increased progressively from 3 weeks to 8 months of age (Figure S2E).

To place these findings in the context of microglia phenotypes in related LSD mouse models, we compared genes upregulated in *Sgsh*^{-/-} microglia with those upregulated in MPS-II^{75–77}/Hunter syndrome⁶⁸ (*lds* deficiency) and mucopolidosis type IV⁷⁴

(B) Cluster proportions across genotypes.

(C) Bulk RNA-seq DEGs (FC > 2, *p*.adj < 0.05) from sorted WT and *Sgsh*^{-/-} microglia (*n* = 3, 4, 2, 2, 4, and 7 mice for 3 week^{WT}, 3 week^{*Sgsh*^{-/-}}, 4 month^{WT}, 4 month^{*Sgsh*^{-/-}}, 8 month^{WT} and 8 month^{*Sgsh*^{-/-}}, respectively).

(D) Heatmap of DEGs significant across all time points in *Sgsh*^{-/-} microglia (*p*.adj < 0.005).

(E) Bar plots of representative repressed, early-upregulated, and late-upregulated genes in *Sgsh*^{-/-} microglia. Data are mean ± standard deviation; *p* values: *p*.adj, DESeq2 (Benjamini-Hochberg).

(F) Pathway enrichment of repressed (FC < -1.5, 8 months [*Sgsh*^{-/-}/WT]), early-upregulated (FC > 2, 3 weeks and 8 months [*Sgsh*^{-/-}/WT]), and late-upregulated genes (FC > 2, 8 months^{*Sgsh*^{-/-}}/3 weeks^{*Sgsh*^{-/-}}). All genes, *p*.adj < 0.05.

(G) Overlap of upregulated microglial genes in LSD models (FC > 1.5, *p*.adj < 0.05): MPS-IIIA (8 months, this study), MPS-II (3 months, Ullman et al.⁶⁸), ML-IV (2 months, Coughnour et al.⁷⁴).

See also Figure S2 and Tables S1 and S3.

(ML-IV,^{78,79} *Mcoln1* deficiency) (Figure 2G). Upregulated genes in *Sgsh*^{−/−} microglia (8 months) overlapped with 62% and 73% of those upregulated in MPS-II (3 months) and ML-IV (2 months), respectively, with a core set of 178 genes shared across all three models (FC > 1.5, *p*.adj < 0.05). Beyond this core set, ML-IV and MPS-II showed little overlap at these time points, indicating microglial responses to lysosomal dysfunction include a shared core program and disease-specific components.

***Sgsh* deficiency primarily affects the epigenetic landscape of microglia**

To investigate mechanisms underlying gene expression changes, we isolated nuclei from major brain cell types⁸⁰ from 8-month-old WT and *Sgsh*^{−/−} mice and performed the assay for transposase-accessible chromatin (ATAC-seq)⁸¹ and histone H3 lysine 27 acetylation (H3K27ac) chromatin immunoprecipitation (ChIP-seq) to profile chromatin accessibility and active regulatory regions (Figure 3A). The resulting datasets (Table S2) were used to integrate reproducible ATAC-seq peaks⁸² with local H3K27ac signal to generate genome-wide, cell type-specific active enhancer/promoter landscapes. Cell type specificity was confirmed at canonical loci (Figure S3A) and by *de novo* motif enrichment analyses of enhancers, which recovered the expected lineage-determining TF (LDTF) motifs for each cell type^{23,83} (Figure S3B).

Comparison of H3K27ac signal at ATAC-seq peaks between WT and *Sgsh*^{−/−} nuclei (|FC| > 1.5, *p*.adj < 0.05) identified ~9,800 differentially acetylated regions in microglia, compared with ~1,000 in oligodendrocytes and 262 in neurons (Figure 3B). The limited changes in oligodendrocytes and neurons parallel their comparatively modest transcriptional response to *Sgsh* deficiency (Figure 1J). To relate these changes to gene expression in *Sgsh*^{−/−} microglia, we plotted the log₂FC of DEGs (8 months, *p*.adj < 0.05) against the log₂FC in H3K27ac at their promoters and nearest enhancers which were also differentially acetylated. This analysis yielded a Pearson correlation of 0.67 (Figure 3C), supporting further analyses of these regions to infer causal relationships. Approximately 90% of ATAC-seq peaks exhibiting altered H3K27ac resided >3 kb from promoters, suggesting the dominant impact of *Sgsh* deficiency on microglial gene expression occurs through distal enhancers.

Examples of genomic regions exhibiting altered chromatin accessibility and H3K27ac in microglia, together with corresponding patterns in oligodendrocytes and neurons, are shown in Figure 3D. *Hpse* is representative of many genes highly and selectively upregulated in *Sgsh*^{−/−} microglia and accompanied by large changes in chromatin accessibility and H3K27ac exclusively in this cell type (Figures 3D and 3E, left). We also observed striking cell-type specificity in epigenetic responses at genes that are broadly expressed but selectively activated in *Sgsh*^{−/−} microglia. For example, *Atp6v1a* is expressed in all three cell types but only upregulated in *Sgsh*^{−/−} microglia, coincident with increased H3K27ac at a putative microglia-specific enhancer (yellow stripe) but not an adjacent putative neuron-specific enhancer (red stripe) (Figures 3D and 3E, middle). Similarly, *Apoe* was expressed in microglia and oligodendrocytes but upregulated only in *Sgsh*^{−/−} microglia, and shared putative enhancers spanning the *Apoe* locus exhibited strongly increased

H3K27ac levels exclusively in microglia (Figures 3D and 3E, right). Collectively, these findings demonstrate a dominant effect of *Sgsh* deficiency on the epigenetic landscape of microglia.

Progressive remodeling of the active chromatin landscape in *Sgsh*-deficient microglia implicates MITF/TFE factors as drivers of altered gene expression

To examine the temporal evolution of active regulatory elements, we performed additional H3K27ac ChIP-seq on WT and *Sgsh*^{−/−} microglia at 3 weeks and 4 months to complement the 8-month data (Figure 4A, left). Applying a >2 FC cutoff (*p*.adj < 0.05) in H3K27ac at ATAC-seq peaks identified 1,896, 5,138, and 5,783 differential enhancer/promoter regions at 3 weeks, 4 months, and 8 months, respectively; the majority of which gained H3K27ac (Figure 4B). Time-dependent decreases and increases in H3K27ac are illustrated at the *Sal1* super-enhancer and *Bhlhe40* loci in Figure 4A, consistent with temporal down- and upregulation of their respective mRNAs (Figure 2E; Table S1). Approximately 76% of ATAC-seq peaks gaining H3K27ac at 3 weeks remained elevated or further increased at 4 and/or 8 months (Figure S4A). Coupled with the >2-fold increase in regions gaining H3K27ac at later time points, these findings indicate progressive activation and expansion of the microglial regulatory landscape during disease progression.

To identify candidate TFs associated with these changes, we performed *de novo* motif enrichment analyses on regions gaining H3K27ac compared with a GC content-matched random genomic background.⁸⁴ The resulting motif classes largely overlapped those in WT microglia but at strikingly different frequencies (Figure 4C). Notably, a motif closely matching the coordinated lysosomal expression and regulation (CLEAR) element preferentially recognized by the MITF/TFE transcriptional regulators was the second most enriched motif at all time points, present in 41%, 31%, and 28% of induced regions (Figure 4C). In contrast, this motif ranked seventh in the active regulatory landscape of WT microglia, occurring in 6.2% of regions (Figure S3B). AP-1 motifs were also enriched at 4 and 8 months, while PU.1 and C/EBP remained prominent across time points consistent with their roles in establishing microglia/macrophage-specific enhancer landscapes. Nuclear factor κB (NF-κB) and interferon regulatory factor (IRF) motifs became modestly enriched at 4 and 8 months, consistent with increased inflammatory gene expression at these stages. In contrast, regions losing H3K27ac showed a more limited motif repertoire dominated by PU.1, MEF2, and SMAD motifs (Figure S4B).

Consistent with these motif results, *Mitf* was upregulated in *Sgsh*^{−/−} microglia at all time points, whereas specific AP-1 (*Atf3*, *Fosb*), IRF (*Irf7*, *Irf8*), and NF-κB (*Nfkb2*, *Relb*) factors were upregulated primarily at 8 months (Figure 4D; Table S1). Many candidate TFs, however, were not differentially expressed, suggesting that altered H3K27ac stemming from *Sgsh* deficiency reflects post-transcriptional regulation of TF activity in addition to transcriptional regulation. Of note, some upregulated TFs (>2-fold), including factors associated with neurodegenerative microglia phenotypes (*Hif1a*, *Rxrg*, and *Bhlhe40*)^{85,86} and *Egr2*, were not recovered by *de novo* motif enrichment analyses but were recovered at lower significance levels when these regions gaining H3K27ac were interrogated against all known motifs in the JASPAR database^{87,88} (Figure S4C). Collectively, these

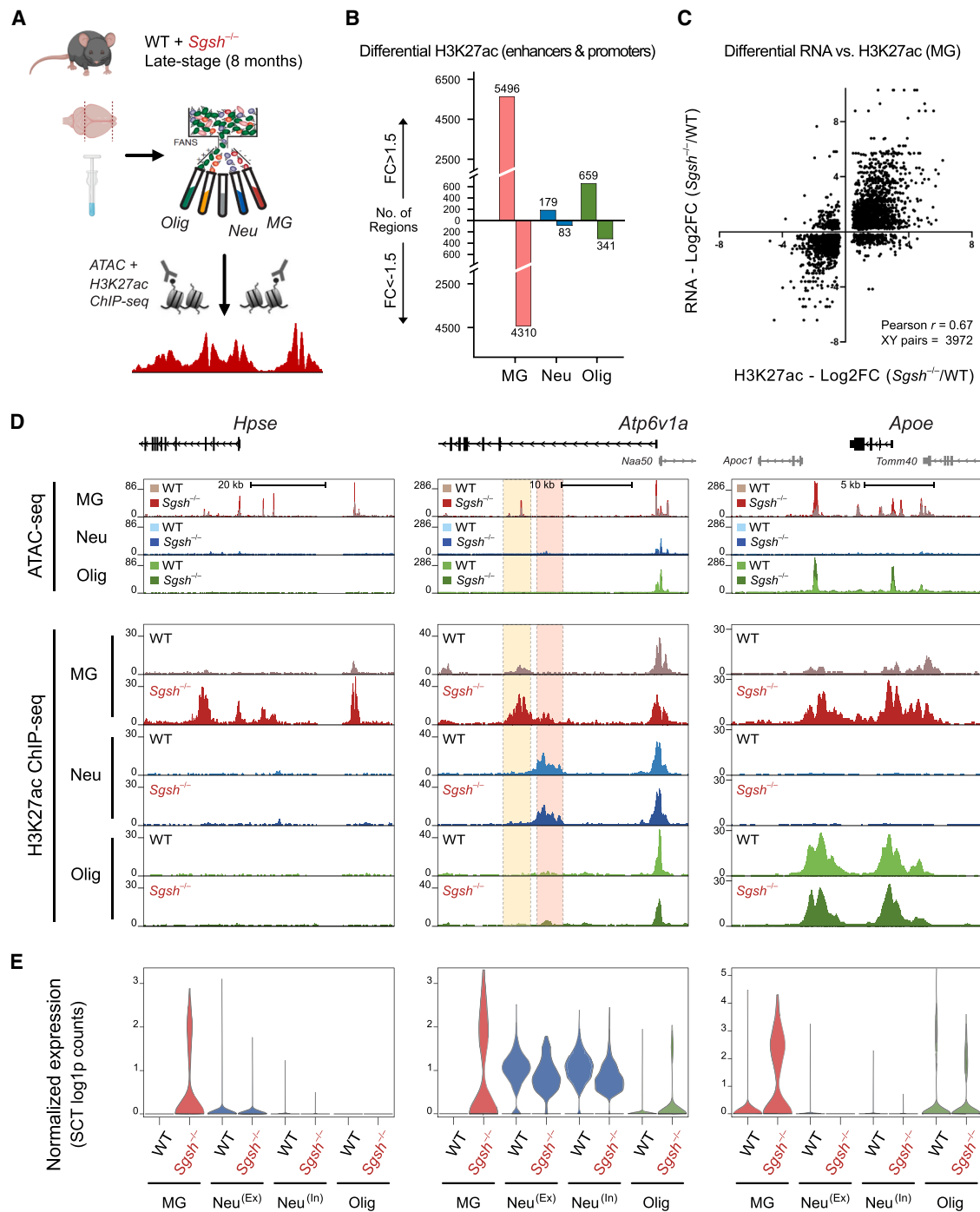


Figure 3. *Sgsh* deficiency primarily affects the epigenetic landscape of microglia

(A) Schematic for brain cell type isolation and downstream assays from 8-month-old mice.

(B) ATAC-seq peaks gaining or losing H3K27ac in microglia (MG), neurons (Neu), and oligodendrocytes (Olig) in *Sgsh*^{-/-} mice.

(C) Scatterplot comparing DEG and H3K27ac log₂FC in *Sgsh*^{-/-} microglia. Data represent DEGs with significant H3K27ac changes (*p*.adj < 0.05) at their nearest ATAC-seq peak.

(D) ATAC-seq and H3K27ac ChIP-seq tracks for WT and *Sgsh*^{-/-} microglia, neurons, and oligodendrocytes at the indicated loci (*n* = 2 [ATAC], *n* = 3 [H3K27ac] mice/genotype in each cell type).

(E) mRNA expression (snRNA-seq) of indicated genes in WT and *Sgsh*^{-/-} microglia, oligodendrocytes and neurons.

See also Figure S3 and Tables S2 and S4.

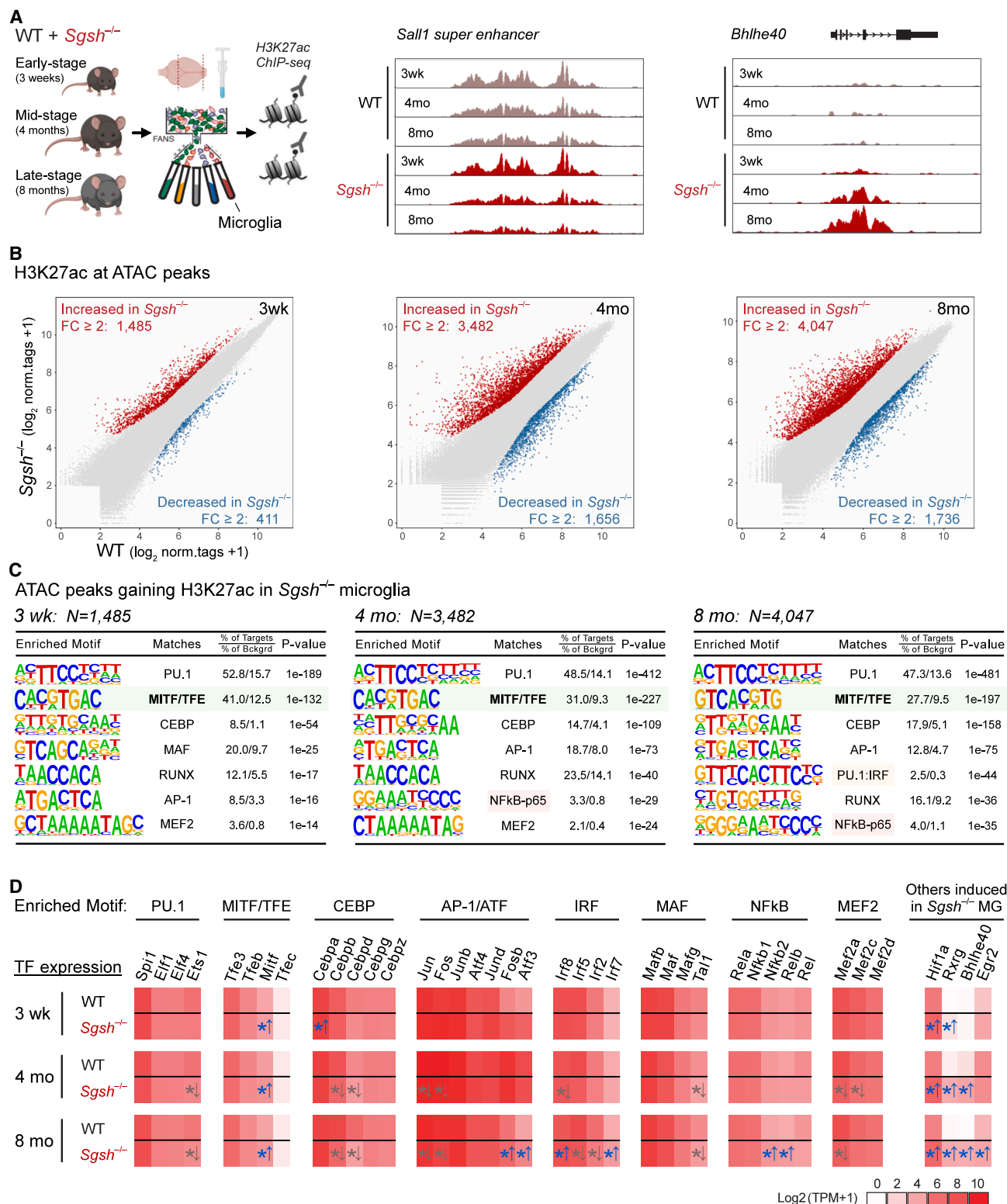


Figure 4. Progressive remodeling of the active chromatin landscape in *Sgsh*-deficient microglia implicates MITF/TFE factors as drivers of altered gene expression

(A) Left: schematic for temporal microglia isolation and downstream assays. Right: H3K27ac tracks at the *Sall1* super-enhancer and *Bhlhe40* loci. (B) H3K27ac tag counts at ATAC-seq peaks in WT and *Sgsh*^{-/-} microglia at the indicated time points. Increased/decreased regions: |FC| > 2, *p*-adj < 0.05.

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findings support a model in which *Sgsh* deficiency results in altered activities of a distinct set of signal-dependent TFs (SDTFs) that cooperate with core microglia LDTFs to drive progressive enhancer remodeling.

Cell type-specific consequences of *Sgsh* deficiency on MITF/TFE factors

The enrichment of a CLEAR-like motif in regions gaining H3K27ac in *Sgsh*^{-/-} microglia, together with progressive *Mitf* upregulation, strongly implicated one or more MITF/TFE family members in the transcriptional response to *Sgsh* deficiency. To investigate the genome-wide binding activities of the MITF/TFE family, we performed ChIP-seq for each expressed factor (MITF, TFEB, and TFE3) in microglia isolated from WT and *Sgsh*^{-/-} mice (4–8 months of age). In parallel, MITF/TFE ChIP-seq assays were performed in oligodendrocytes, the second major brain cell type with significant expression of *Mitf*, *Tfeb*, and *Tfe3*. Following irreproducibility discovery rate (IDR) analyses, high-quality peak sets were obtained for MITF/TFE factors in WT and *Sgsh*^{-/-} microglia and oligodendrocytes.

Representative MITF, TFEB, and TFE3 binding profiles in microglia and oligodendrocytes at promoters and putative enhancers associated with *Olig1*, *Apoe*, and *Hpse* are shown in Figure 5A. Genome-wide, thousands of reproducible binding sites were identified in each cell type and genotype (Figures S5A and S5B). *Sgsh* deficiency increased binding of all three factors at most genomic binding sites in both cell types, consistent with their nuclear translocation commonly observed during conditions of cellular and/or lysosomal stress.^{89,90} However, high-magnitude changes in binding (FC > 8) were observed almost exclusively in microglia (Figures S5C and S5D). In microglia, MITF and TFEB binding overlapped nearly completely with the larger TFE3 cistrome (Figure S5E), and a similar, although less extensive, pattern was observed in oligodendrocytes (Figure S5F), consistent with their binding as exclusive hetero- and homodimers.^{91,92}

As TFE3 encompassed the majority of MITF and TFEB binding sites in both cell types, we utilized its cistrome as a representative readout of MITF/TFE binding. Comparison of TFE3 binding profiles in microglia and oligodendrocytes revealed moderate overlap at promoters (Figure 5B), where shared sites were enriched for motifs recognized by promoter-biased TFs (e.g., SP1 and NFY) together with the CLEAR motif (Figure 5C). The corresponding genes were enriched for functions related to macromolecule metabolism, endosomal transport, and cellular responses to stress; processes broadly consistent with established MITF/TFE functions (Figure 5D). In contrast, TFE3 binding at putative enhancers was largely cell type-specific (Figure 5E). For example, at the *Olig1* locus, MITF, TFEB, and TFE3 binding was observed exclusively at putative oligodendrocyte-specific enhancers and changed little with *Sgsh* deficiency (Figure 5A, left). Conversely, at the *Hpse* locus, MITF/TFE binding was highly induced at regulatory elements selectively active in microglia (Figure 5A, right). Consistent with this, TFE3-bound

regions at microglia enhancers were enriched for the CLEAR element together with motifs recognized by microglia LDTFs (Figure 5F), with the nearest expressed genes showing enrichment for functional annotations linked to immune cell function and lysosomal-related processes (Figure 5G). Conversely, TFE3-bound regions at oligodendrocyte enhancers were enriched for the CLEAR element together with motifs recognized by oligodendrocyte LDTFs (Figure 5H), with the nearest expressed genes showing enrichment for terms linked to oligodendrocyte function, including ensheathment of neurons (Figure 5I). Together, these findings suggest that MITF/TFE factors exert shared functions in microglia and oligodendrocytes while also contributing to cell-specific programs through collaborative interactions with LDTFs at distal regulatory elements.

Association of MITF/TFE factors with homeostatic and stress-responsive transcriptional states

The acquisition of genome-wide MITF/TFE binding profiles in WT and *Sgsh*^{-/-} microglia enabled direct evaluation of how these factors were associated with altered epigenetic landscapes. Two general categories of binding events were observed: sites exhibiting basal occupancy in WT microglia that remained stable or decreased with *Sgsh* deficiency, and sites with low/basal binding that significantly increased with *Sgsh* deficiency. An example of the first category is the intronic enhancer region of *Hexb* (Figure 6A, yellow stripe). This peak lies near a region showing variable increases of MITF/TFE binding (blue stripe); however, H3K27ac across the *Hexb* region was unchanged and *Hexb* expression was unaltered in *Sgsh*^{-/-} microglia. In contrast, at the *Cst7* locus, MITF and TFE3 binding increased markedly in *Sgsh*^{-/-} microglia (Figure 6B, yellow stripe), coincident with increased H3K27ac and strong upregulation of *Cst7* mRNA. A similar pattern was observed at *Hpse* (Figure 5A), which was also strongly upregulated in *Sgsh*^{-/-} microglia.

To examine these relationships genome-wide, we analyzed MITF, TFEB, and TFE3 tag density at ATAC-seq peaks exhibiting altered H3K27ac in *Sgsh*^{-/-} microglia. Regions with >2-fold increases in H3K27ac showed strong increases in MITF/TFE binding (Figures 6C and 6D, 1st row), consistent with the patterns at *Cst7* and *Hpse*. Conversely, regions with >2-fold decreases in H3K27ac exhibited reduced MITF/TFE occupancy (Figure 6D, 2nd row). To investigate this pattern at direct MITF/TFE binding sites, we isolated regions with differential factor binding and quantified H3K27ac. Here, regions with increased MITF/TFE binding showed significantly elevated local H3K27ac, whereas regions with reduced binding exhibited decreased H3K27ac (Figure 6D, 3rd and 4th rows). Additionally, at matched genomic regions, larger increases in MITF/TFE occupancy correlated with higher levels of H3K27ac (Figure S6A). Together, these findings are consistent with MITF/TFE factors contributing to enhancer activation at a large fraction of regulatory regions affected by *Sgsh* deficiency.

However, increased MITF/TFE binding was not invariably associated with enhancer activation, as a subset of loci gained

(C) *De novo* motif enrichment of ATAC-seq peaks gaining H3K27ac (FC > 2, *p*.adj < 0.05) in *Sgsh*^{-/-} microglia from (B).

(D) Heatmap of the most highly expressed TFs recognizing enriched motifs in (C). Asterisks, significant differential expression in *Sgsh*^{-/-} microglia (|FC| > 1.35, *p*.adj < 0.05); arrows, direction of change.

See also Figure S4 and Tables S1 and S2.

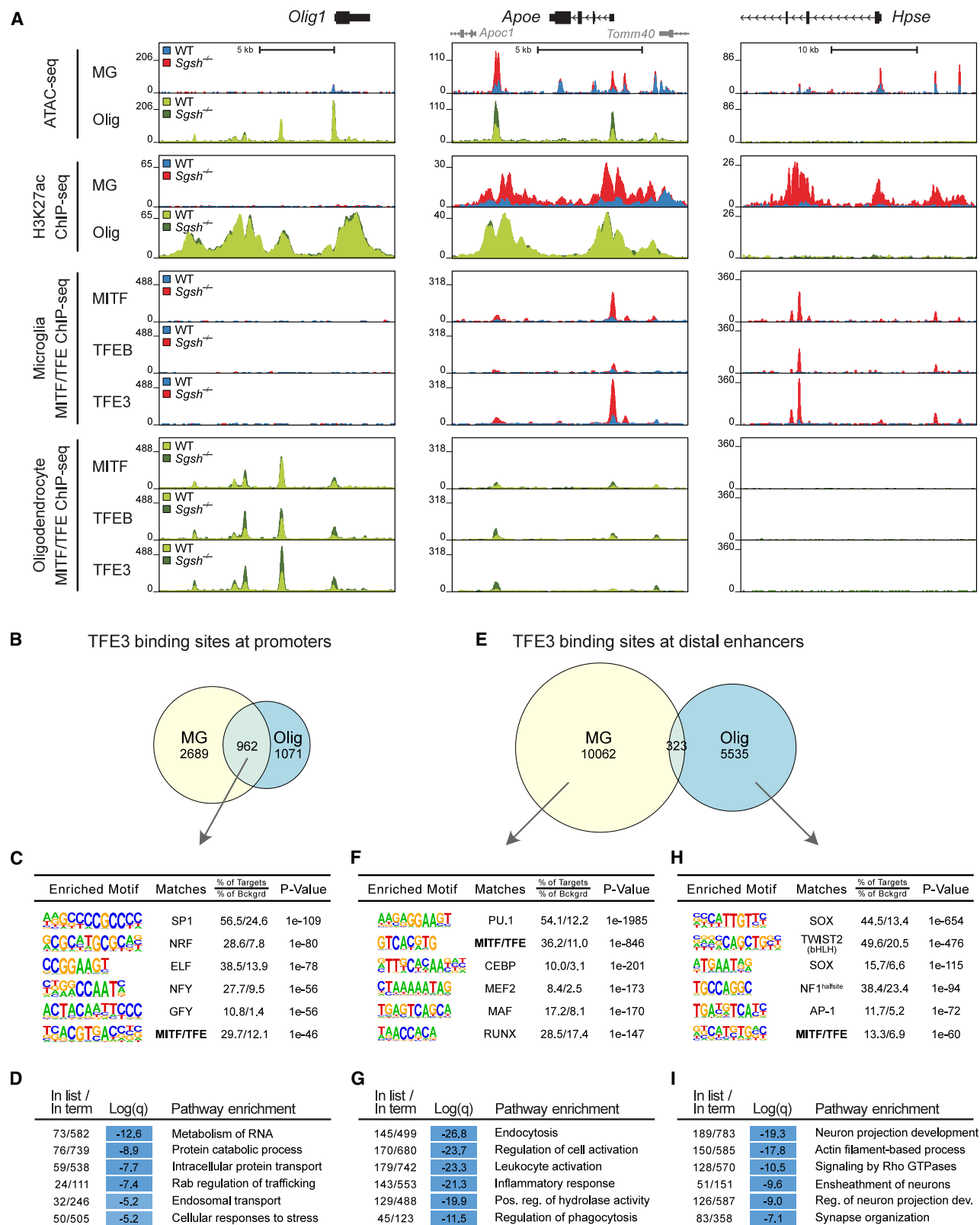


Figure 5. Cell type-specific consequences of *Sgsh* deficiency on MITF/TFE factors

(A) ATAC-seq and ChIP-seq tracks for H3K27ac, MITF, TFEB, and TFE3 in WT and *Sgsh*^{-/-} microglia (MG) and oligodendrocytes (Olig) at the indicated loci (*n* = 2 [ATAC], *n* = 3 [H3K27ac] mice/genotype in each cell type).

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MITF/TFE binding without increases in H3K27ac or nearby gene expression. To investigate the basis for this observation, we compared strongly bound TFE3 sites (>64 normalized tags) that gained H3K27ac (>2-fold) in *Sgsh*^{-/-} microglia with similarly bound sites in which H3K27ac was unchanged (<1.2 FC). At TFE3 binding sites with increased H3K27ac, a consensus MITF/TFE binding site was the most enriched motif, present in 62% of regions, followed by motifs for PU.1, C/EBP, AP-1, and TFAP4/HLH factors (Figure 6E). In contrast, among TFE3 binding sites with unchanged H3K27ac, the MITF/TFE motif was less enriched and only present in 30.5% of regions (Figure S6B). This latter set was additionally enriched for GC-rich motifs (SP1, Krüppel-like factors [KLFs]) and a motif recognized by MAF factors. Of note, although NF-κB motifs became moderately enriched in ATAC-seq peaks gaining H3K27ac at 4 and 8 months (Figure 4C), they were not generally enriched in peaks strongly bound by TFE3 (Figure 6E).

To gain insights into the basis for the locus-specific changes in H3K27ac in relation to TFE3 binding, we analyzed co-occurrence frequencies of the top motifs over a range of motif thresholds.⁹³ At an intermediate threshold, 37 of 128 possible motif combinations exhibited significant bias toward one peak class or the other, with the 20 most significant (*p* < 0.002) shown in Figure 6F. Notably, all combinations associated with increased H3K27ac contained a consensus MITF/TFE motif, whereas only 3 of 9 combinations associated with unchanged H3K27ac did so. Motif combinations including AP-1, C/EBP, and ATF factors were strongly associated with peaks gaining H3K27ac, whereas combinations including SP1/KLF motifs were biased toward peaks exhibiting unchanged H3K27ac. This general pattern was robustly observed across motif thresholds, while PU.1 and MEF2 motifs contributed little to differential peak bias. These analyses suggest that both the presence of a MITF/TFE motif and combinations of nearby motifs for additional factors (i.e., C/EBP and AP-1/ATF) are required for MITF/TFE binding to result in productive enhancer activation.

Molecular consequences of MITF + TFE3 depletion in *Sgsh* deficiency

To investigate direct transcriptional functions of MITF/TFE factors *in vivo*, we designed antisense oligonucleotides (ASOs) targeting mouse *Mitf*, *Tfeb*, and *Tfe3*. Single bolus intracerebroventricular administration into 3-month-old WT mice produced efficient, target-specific reduction in gene expression in microglia isolated 14 days later (Figures S6C and S6D). Despite widespread MITF/TFE binding in microglia, individual factor mRNA reduction in WT mice resulted in relatively few global gene expression changes, consistent with functional redundancy within the family^{90,94} and stress-dependent activity. We therefore performed combined *Mitf* and *Tfe3* ASO administration experiments in *Sgsh*^{-/-} mice (Figure S6E), selecting these factors

because they occupied the greatest number of genomic sites in microglia (Figure S5A) and could be co-targeted within dosing constraints. Combined ASO treatment reduced *Mitf* and *Tfe3* mRNA by 80% and 56%, respectively (Figure S6F), and resulted in widespread, lower magnitude changes in microglial gene expression (444 downregulated, 482 upregulated) relative to control ASO-treated *Sgsh*^{-/-} mice (*p*.adj < 0.05) (Figure 6G).

Genes downregulated by *Mitf* + *Tfe3* depletion significantly overlapped with genes upregulated in *Sgsh*^{-/-} microglia, reflected by a negative correlation between datasets (Figure 6H, bottom; Spearman *ρ* = -0.30). This set included known MITF/TFE-regulated genes identified *in vitro* (e.g., *Hif1a*, *Gpnmb*, and *Clec7a*),^{95,96} as well as additional candidates including *Cst7*, *Itgax*, and *Rxrg* (Figure 6H, top). Notably, more than one-third of the ~200 most significant DEGs upregulated in *Sgsh*^{-/-} microglia across time points were downregulated by *Mitf* + *Tfe3* ASO treatment (Figure S6G). Consistent with this, downregulated genes were enriched for functional annotations highly similar to those enriched in *Sgsh*^{-/-} upregulated genes, including lysosome, neutrophil degranulation, and regulation of wound healing (Figure 6I). In contrast, genes upregulated by *Mitf* + *Tfe3* depletion were enriched for interferon-responsive and MHC class II programs, including *Oas2*, *Ifit1/2/3*, *Isg15*, and *Irf7*, many of which showed moderate upregulation in *Sgsh*^{-/-} microglia (Figures 6J and 2F). This pattern is consistent with recent evidence that MITF/TFE family members can limit stimulator of interferon genes (STING)-dependent type I interferon outputs in macrophages.⁹⁷ Collectively, these reciprocal effects indicate that MITF/TFE activity underlies a substantial component of the *Sgsh*^{-/-} transcriptional program, promoting expression of lysosomal and other disease-activated genes while constraining a distinct immune-associated module.

The microglial *Sgsh*^{-/-} signature is observed in mouse models of age-related neurodegeneration and in AD patients

To examine the relevance of these findings to age-related neurodegenerative diseases, we intersected genes upregulated in *Sgsh*^{-/-} microglia with those upregulated in two mouse models: (1) amyloid-responsive, *Clec7a*^{high} microglia (AR-MG) from APP/PS1 mice⁷³ and (2) tau/APOE4-responsive microglia (TERM) from mice co-expressing human P301S tau and APOE4 (TE4)⁹⁸ (Figure 7A, left). AR-MG, which possess a MGnD/DAM phenotype, are linked to TREM2-dependent protective responses to amyloid pathology,^{72,99,100} whereas TERM are associated with the severe neurodegeneration present in TE4 mice.^{98,101,102} Using a FC > 1.5 (*p*.adj < 0.05), we defined 1,599 and 1,381 upregulated genes in AR-MG and TERM, respectively. Of the 358 genes upregulated >1.5-fold (*p*.adj < 0.001) in *Sgsh*^{-/-} microglia at 3 weeks, 205 (57%) were similarly upregulated in AR-MG and 166 (46%) shared with TERM, with 125 shared across the

(B) Overlap of active TFE3 binding sites at promoters (TSS ± 500 bp) in *Sgsh*^{-/-} microglia and oligodendrocytes.

(C and D) *De novo* motif and pathway enrichment at TFE3-bound promoters shared between microglia and oligodendrocytes.

(E) Overlap of active TFE3 binding sites at distal genomic regions (>500 bp from TSS) in *Sgsh*^{-/-} microglia and oligodendrocytes.

(F and G) *De novo* motif and nearest-gene pathway enrichment at microglia-specific distal TFE3 binding sites.

(H and I) *De novo* motif and nearest-gene pathway enrichment at oligodendrocyte-specific distal TFE3 binding sites.

Active sites: TFE3 IDR peaks with >64 normalized merged tags and >16 H3K27ac.

See also Figure S5 and Table S2.

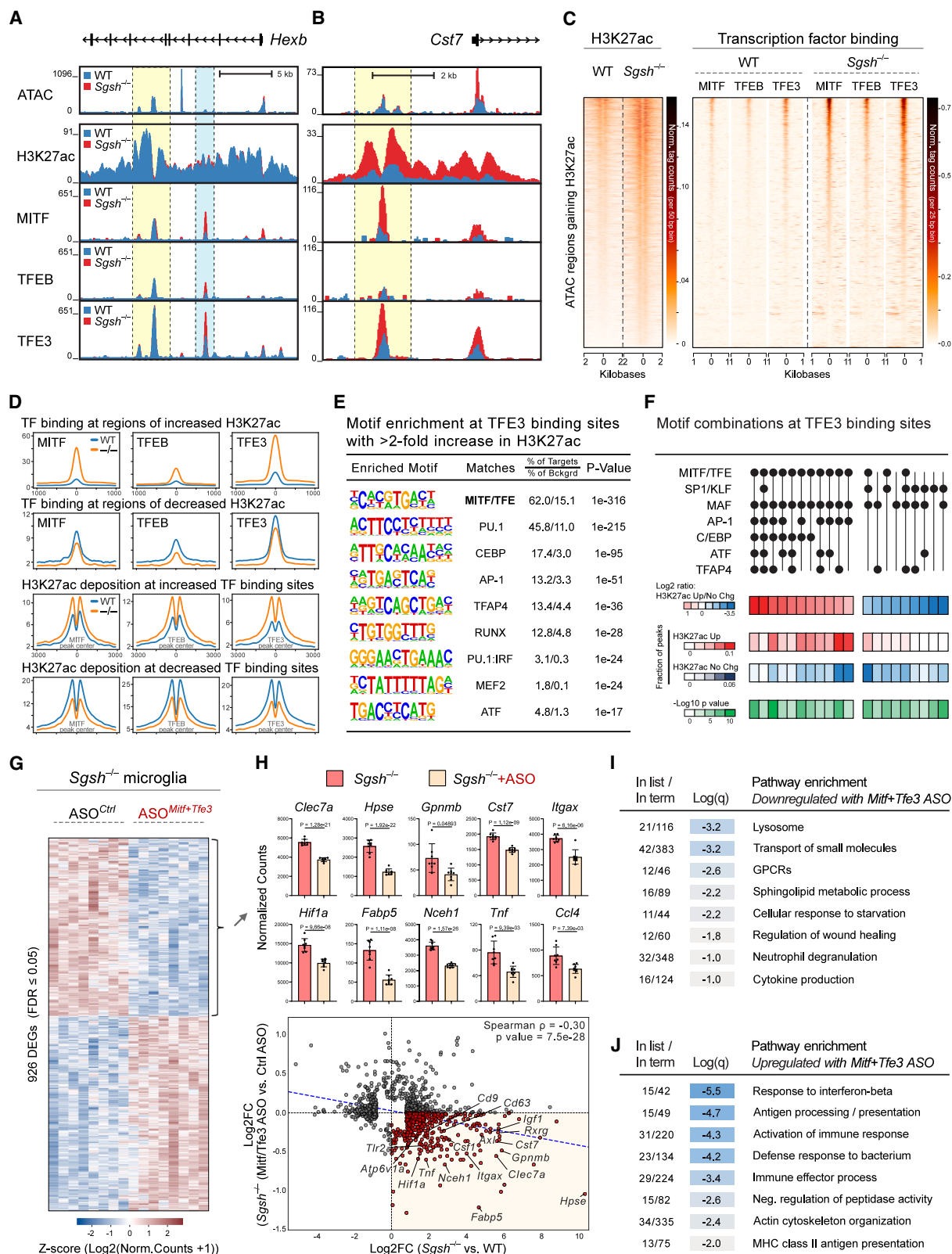


Figure 6. Association of MITF/TFE factors with homeostatic and stress-responsive states and molecular consequences of their depletion in *Sgsh* deficiency

(A and B) ATAC-seq and ChIP-seq tracks for H3K27ac and MITF/TFE factors in WT and *Sgsh*^{-/-} microglia at the indicated loci.

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datasets (Figure S7A, left). This overlap expanded at 8 months, with 669/1,162 genes (58%) in *Sgsh*^{-/-} microglia similarly upregulated in AR-MG (42% of all AR-MG genes), and 406 (35%) shared with TERM (30% of all TERM genes). 270 genes were shared across all three models, and as a group showed strong enrichment for lysosomal processes (Figure 7A, right). Notably, ~one-third of these 270 shared genes were downregulated in microglia following *Mitf* + *Tfe3* ASO administration in *Sgsh*^{-/-} mice (Figure 7B), linking this shared response to MITF/TFE activity.

Apart from this overlap, each model showed program-specific features. Genes uniquely upregulated in *Sgsh*^{-/-} microglia were enriched for pathways including response to type II interferon, antigen processing/presentation, and extracellular matrix organization, reflecting heightened interferon and lysosome-ECM dysregulation characteristic of LSDs^{104–106} (Figure S7B, right). *Sgsh*^{-/-} microglia also exhibited the most upregulated lysosomal genes among the three models (Figure S7C), consistent with their pronounced lysosomal defect. In contrast, AR-MG-specific genes were enriched for metabolic and mitochondrial function pathways consistent with altered microglial energetic metabolism in AD-like amyloid pathology¹⁰⁷ (Figure S7B, left). Lastly, TERM-specific genes were enriched for pathways including dendrite development, synaptic vesicle processing, and regulation of neuron apoptotic process, aligning with the marked neurodegeneration in the TE4 model as compared with the other two models (Figure S7B, middle).

To extend these comparisons to the epigenetic level, we analyzed ATAC-seq profiles from AR-MG¹⁰⁸ and compared them with *Sgsh*^{-/-} microglia. Although ATAC-seq is less sensitive than H3K27ac in detecting regulatory changes, we identified 1,842 regions with increased accessibility in AR-MG and 2,403 in *Sgsh*^{-/-} microglia compared with control microglia (FC > 2.5, *p*.adj < 0.1). Regions of increased accessibility in the two models showed notable overlap, with 420 shared peaks, alongside substantial model-specific peaks (Figure 7C). Shared induced regions, aligned with ChIP-seq data from *Sgsh*^{-/-} and WT mice, are illustrated at the *Nceh1* and *Bhlhe40* loci in Figure 7D. Notably, a subset of these regions exhibited large increases in MITF and TFE3 binding in *Sgsh*^{-/-} microglia. Consistent with this, expression of their nearest genes was often reduced in *Sgsh*^{-/-} mice treated with *Mitf* + *Tfe3* ASOs (Figures 7B and 6G). Collectively, these findings highlight common regulatory mechanisms contributing to chromatin and transcriptional changes in *Sgsh*-deficient and amyloid pathology mouse models.

Finally, to investigate whether the *Sgsh*^{-/-} transcriptional program is present in human disease, we analyzed two large human

microglia datasets from prefrontal cortex: (1) fresh biopsies from 52 individuals (Gazestani et al.¹⁰³ and (2) postmortem samples from 437 individuals (Green et al.),¹⁰⁹ together spanning the AD continuum as well as individuals without clinical or neuropathological features of AD. Using the ~350 most significant upregulated genes in *Sgsh*^{-/-} microglia as a signature, we calculated aggregated enrichment Z scores across the human microglia populations. In the fresh biopsy cohort, the *Sgsh*^{-/-} signature was highly enriched in three GPNMB-high¹¹⁰ clusters collectively designated GPNMB-LPL (Figures 7E, 7F, and S7D), a population uniquely associated with AD pathology in this study. In the postmortem dataset, the *Sgsh*^{-/-} signature was similarly enriched in the strongest pathology-associated microglial clusters (i.e., Mic.12/Mic.13; lipid-associated), as well as reactive redox, inflammatory, and interferon-response clusters (Figures S7E and S7F). Together, these analyses indicate that the microglial response to lysosomal dysfunction resulting from *Sgsh* deficiency significantly overlaps with mouse and human microglial responses to AD pathology.

DISCUSSION

In contrast to common and sporadic forms of age-related neurodegenerative diseases, in which altered transcriptional states are concentrated in subsets of cells that are localized to focal areas of pathology, germline *Sgsh* deficiency has the potential to exert cell-autonomous consequences in all cell types in which it is expressed. Despite this potential, our ultrastructural, immunohistochemical, transcriptomic, and epigenetic analyses demonstrate a disproportionate effect of *Sgsh* deficiency on microglia. The morphological changes suggest microglia may play major roles in uptake and clearance of extracellular SGSH substrates (i.e., HS) in the brain, although lysosomal changes can arise from other factors and do not alone confirm this function. Together with the relatively modest changes in gene expression and morphology of other brain cell types, the marked gene expression changes and lysosome expansion in microglia are consistent with a potential compensatory function of microglia early in disease, which may serve to mitigate the pathogenic effects of HS accumulation in other brain cell types.

This interpretation is supported by recent studies of microglia in other LSDs. In Niemann-Pick disease models, microglial depletion significantly exacerbates brain pathology, indicating a protective role which becomes corrupted upon their lysosomal overload.¹¹¹ In Sandhoff disease (*Hexb*^{-/-}) models, microglia were shown to be a principal source of neuronal HEXB, which

(C) Normalized tag density distributions at ATAC-seq peaks gaining H3K27ac (FC > 2) in *Sgsh*^{-/-} microglia (4 months). Left: H3K27ac; right: MITF/TFE factors at the same regions. H3K27ac: 50-bp bins, scaled to 500 tags/4 kb region. TFs: 25-bp bins, scaled to 750 tags/2 kb region.

(D) Rows 1–2, average MITF/TFEB/TFE3 signal at regions of increased and decreased H3K27ac ($\log_2\text{FC} > 1$ or < -1 , *p*.adj < 0.05, 4 months). Rows 3–4, average H3K27ac signal (4 months) at regions of increased and decreased MITF/TFEB/TFE3 binding ($\log_2\text{FC} > 1$ or < -1 , *p* < 0.05).

(E) *De novo* motif enrichment at ATAC-seq peaks with >64 normalized TFE3 tags and >2-fold increase in H3K27ac in *Sgsh*^{-/-} microglia.

(F) Motif co-occurrence analysis of TFE3 binding sites with increased (left) or unchanged (right) H3K27ac in *Sgsh*^{-/-} microglia.

(G) Heatmap of microglial DEGs (*p*.adj < 0.05) from 3-month-old *Sgsh*^{-/-} mice treated with *Mitf* + *Tfe3*-targeting ASOs or control ASO (*n* = 8 mice/treatment group).

(H) Top: example downregulated genes from (G). All genes significantly upregulated in *Sgsh*^{-/-} microglia versus WT (*p*.adj < 0.05). Bottom: $\log_2\text{FC}$ comparison of *Sgsh*^{-/-} DEGs ($|\log_2\text{FC}| > 0.9$, *p*.adj < 0.001) and DEGs following *Mitf* + *Tfe3* ASO administration in *Sgsh*^{-/-} microglia (*p*.adj < 0.001). Data represent genes meeting significance criteria in either comparison (*n* = 1,100).

(I and J) Pathway enrichment of downregulated (I) or upregulated (J) genes in *Sgsh*^{-/-} microglia after ASO-mediated *Mitf* + *Tfe3* depletion *in vivo*. See also Figure S6.

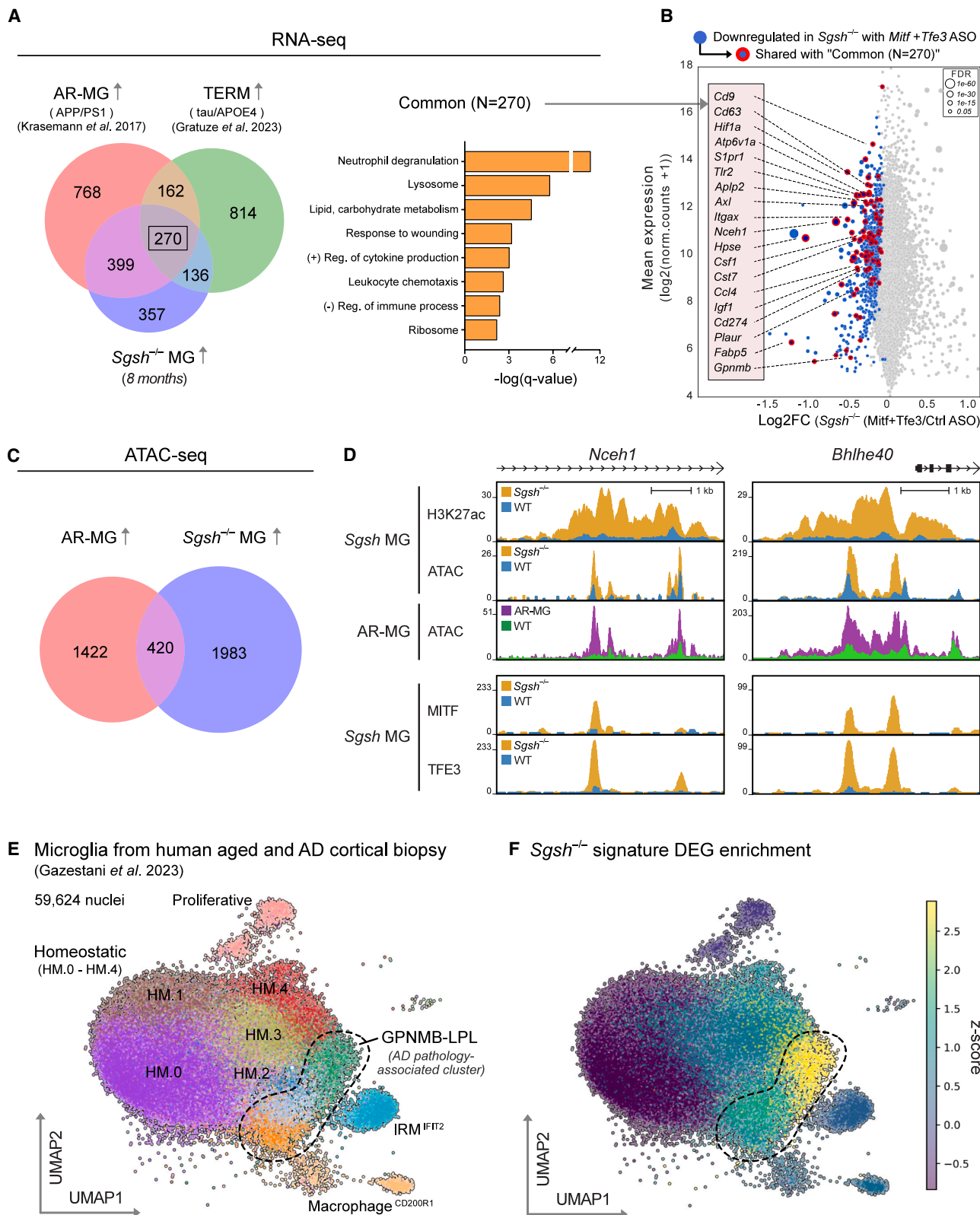


Figure 7. The microglial $Sgsh^{-/-}$ signature is observed in mouse models of age-related neurodegeneration and in AD patients

(A) Left: overlap of genes upregulated >1.5-fold in 8-month-old $Sgsh^{-/-}$ microglia ($p_{\text{adj}} < 0.001$), *Clec7a*^{high} AR-MG (APP/PS1 mice [9 months]) and TERM (TE4 mice) ($p_{\text{adj}} < 0.05$). Right: pathway enrichment of commonly upregulated genes.

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neurons take up to support clearance of neurotoxic GM2 gangliosides. Accordingly, replacing *Hexb*^{-/-} microglia with *Hexb*-sufficient cells reduces substrate accumulation and neuroinflammation and improves neuronal and motor outcomes.^{112,113} Further, in MPS-IIIA models, hematopoietic stem cell gene therapy is effective primarily when *SGSH* expression is driven by a myeloid/microglial-specific promoter, highlighting the cell type's dominant role in disease rescue.¹¹⁴ We speculate that neuropathology in MPS-IIIA ensues when the capacity of microglia and their lysosomes to perform this initial function eventually becomes overwhelmed, resulting in a combination of increased cell-autonomous effects of HS accumulation in neurons and other brain cell types and neurotoxic effects of the altered program of gene expression in microglia. The transition in functional annotations of differentially regulated genes, showing progressive decreases in genes associated with homeostatic microglia functions and an increasingly inflammatory signature from 3 weeks to 8 months, is consistent with this possibility. Based on this interpretation, early administration of small molecules,¹¹⁵ cell replacement therapies in which *SGSH*-mutant microglia are replaced with WT microglia, or emerging brain-penetrant enzyme replacement therapies have potential for therapeutic benefit in disorders affecting microglia or multiple CNS cell types.^{68,116}

Quantitative analysis of chromatin accessibility and H3K27ac identified substantial changes in *Sgsh*^{-/-} microglia, but significantly fewer changes in neurons or oligodendrocytes, consistent with gene expression data. Epigenetic changes in microglia as defined by increased H3K27ac provide evidence for dominant and context-dependent roles of members of the MITF/TFE family as major drivers of the transcriptional consequences of *Sgsh* deficiency that are dependent on collaborative interactions with AP-1/ATF, C/EBP, and PU.1/ETS TFs. It will therefore be of interest to expand analysis of epigenetic changes in other LSDs and in contexts in which lysosomes are dysfunctional and/or damaged.

Although roles of MITF/TFE factors in the regulation of lysosomal function, autophagy, inflammation, and the cellular stress response have been studied in depth,^{89,90,117–121} nearly all molecular analyses of these functions have been performed using *in vitro* model systems. While TFE3 binding has been recently profiled *in vivo* in bulk mouse liver demonstrating altered binding at the *Insig2* locus as a consequence of reduced mTORC1 signaling,¹²² to our knowledge, the genome-wide locations of the MITF/TFE factors have not been systematically determined in a cell type-resolved manner *in vivo*, under either homeostatic or disease conditions. Here, ChIP-seq analyses of MITF, TFEB, and TFE3 in microglia and oligodendrocytes demonstrated a high degree of overlap in the binding of these factors within

each cell type but highly divergent binding patterns across cell types, particularly at putative distal microglia- or oligodendrocyte-specific enhancers. At these locations, motif enrichment analysis indicated very similar MITF/TFE motifs in both cell types but different co-enriched motifs that represented DNA recognition elements for microglia-specific or oligodendrocyte-specific LDTFs, respectively. These findings are consistent with a model in which the genomic binding sites for MITF/TFE factors are determined in each cell type by collaborative interactions with combinations of microglia or oligodendrocyte-specific LDTFs and additional more broadly expressed SDTFs, which collectively enable cell-specific signal-dependent effects on gene expression. Consequently, these findings provide evidence that MITF/TFE factors likely contribute to cell type-specific biological functions in microglia and oligodendrocytes beyond their shared roles in regulation of lysosomal biology.

The binding patterns of MITF, TFE3, and TFEB at genomic regions exhibiting increased H3K27ac in microglia are consistent with these factors directly contributing to a large fraction of the epigenetic and transcriptional changes resulting from *Sgsh* deficiency. The mechanisms by which loss of function of *Sgsh* leads to activation of MITF/TFE and presumably AP-1/ATF factors remain to be established. Unexpectedly, in the case of oligodendrocytes, sites of increased binding of MITF/TFE factors were less often associated with increased H3K27ac and nearby gene expression. In addition, increased binding of MITF/TFE factors in microglia was not invariably associated with increased H3K27ac and nearby gene expression. These observations suggest that an important additional layer of regulation beyond nuclear translocation and sequence-specific DNA binding is required for gene activation that cannot be explained by current models. Motif co-occurrence analysis suggests that this additional layer may be determined at least in part by specific combinations of co-binding factors. These findings thereby reveal unexpected complexities in gene activation by MITF/TFE factors that may be linked to both protective and pathological functions.

It was of particular interest to observe the overlap of upregulated genes in microglia with *Sgsh* deficiency with those that are upregulated in microglia responding to amyloid- or tau/APOE4-induced pathology, a relationship that was also observed at the level of altered chromatin accessibility in the amyloid model. At these locations, there was a high degree of overlap with binding sites for MITF/TFE factors from *Sgsh*^{-/-} microglia, providing evidence that mechanisms underlying the transcriptional response to lysosomal stress in *Sgsh* deficiency may also contribute to the altered epigenetic landscapes and gene expression programs of microglia in amyloid and related neurodegenerative models. Consistent with this, a recent study demonstrated that the TFEB-vATPase signaling axis, a key

(B) MA plot highlighting *Mitf* + *Tfe3* ASO-downregulated genes in *Sgsh*^{-/-} microglia (blue; *p*-adj < 0.1), with red indicating those overlapping the commonly upregulated genes across models in (A) (82/270 genes). Pink box, example genes.

(C) Overlap of ATAC-seq peaks gaining accessibility in *Sgsh*^{-/-} microglia and AR-MG (9 months).

(D) ATAC-seq and ChIP-seq tracks at the *Nceh1* and *Bhlhe40* loci in *Sgsh*^{-/-} and AR-MG microglia, and corresponding WT controls.

(E) UMAP of microglia/macrophages from human cortical biopsies (replotted from Gazestani et al.¹⁰³). The GPNMB-LPL population includes GPNMB-NACA (green), GPNMB-EYA2 (orange) and LPL-CD83 (light blue) subclusters. Each dot, one cell.

(F) Gazestani et al. UMAP colored by enrichment of the *Sgsh*^{-/-} signature DEG set (human orthologs). Values reflect cluster-mean scores *Z* scored across all clusters in (E).

See also Figure S7.

regulator of lysosomal function, plays a critical role in microglial activation and dysfunction in a tauopathy mouse model, further supporting a role for MITF/TFE-mediated pathways in age-related neurodegenerative diseases.¹²³ A major question for future investigation is whether this response serves neuroprotective and/or pathogenic roles. To the extent that activation of the MITF/TFE family of TFs enhances the ability of lysosomes to execute their normal functions and compensate for perturbation-associated neuropathology, the functions of these factors are likely to be protective. However, the collateral activation of genes associated with inflammation is likely pathogenic. An important objective for the future will be to determine whether the context- and gene-specific activities of MITF/TFE factors can be influenced for therapeutic purposes.

Limitations of the study

Although EM analyses of frontal cortex indicate that the most prominent lysosomal abnormalities occur in microglia, we cannot exclude important transcriptomic or epigenetic changes in astrocytes due to low recovery. Likewise, neuronal epigenetic alterations may be restricted to specific subpopulations and therefore diluted in bulk NeuN⁺ analyses. *In vivo* ASO-mediated depletion of *Mitf* and *Tfe3* could affect gene expression in microglia through cell-autonomous and non-cell-autonomous mechanisms. Lastly, partial functional redundancy of MITF, TFEB, and TFE3 results in only partial loss of function of these factors in the context of ASO experiments, limiting the effects of these perturbations on cellular phenotypes beyond changes in gene expression.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources/reagents should be directed to and will be fulfilled by the lead contact, Christopher K. Glass (ckg@ucsd.edu).

Materials availability

This study did not generate new, unique reagents.

Data and code availability

UCSC browser tracks for RNA-seq, ATAC-seq, and ChIP-seq data can be accessed via the following link: https://genome.ucsc.edu/s/cbalak/Balak_et_al_2024_SgshKO. All raw and processed bulk sorted RNA-seq, sn/scRNA-seq, ATAC-seq, and ChIP-seq data are available via the NCBI GEO repository under accession numbers GEO: GSE278447, GSE278448, GSE278449, and GSE278451. This paper does not report original code. Any additional information required to reanalyze data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, C.D.B., J.C.M.S., and C.K.G.; investigation, C.D.B., J.C.M.S., A.J.L., E.W., and N.J.S. assisted by M.P.P., S.O., and J.A.B.; *Sgsh*^{−/−} mice, P.L.S.M.G.; ASO development, validation, and WT injections, C.H., J.D., and F.K.; immunofluorescent imaging/analyses, B.L. and Y.Z.; motif co-occurrence analyses, P.S. and C.K.G.; *Sgsh*^{−/−} signature enrichments, V.S., C.D.B., and B.S.; writing (original draft), C.D.B., J.C.M.S., and C.K.G.; writing (review/editing), all authors.; supervision, C.K.G.

DECLARATION OF INTERESTS

C.K.G. is a cofounder and member of the scientific advisory board of Asteroid Therapeutics. C.H., J.D., and F.K. are employees and/or shareholders of Ionis Pharmaceuticals, Inc.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-mouse CD16/32	BioLegend	101319; RRID: AB_1574973
APC anti-mouse/human CD11b	BioLegend	101212; RRID: AB_312794
Alexa Fluor 488 anti-mouse CD45	BioLegend	103122; RRID: AB_493531
PE anti-mouse CX3CR1	BioLegend	149006; RRID: AB_2564314
Rabbit anti-mouse/human PU.1 PE (9G7)	Cell Signaling	81886S; RRID:AB_2799984
Mouse anti-mouse/human NeuN AF488 (A60)	MilliporeSigma	MAB377X; RRID: AB_2149209
Rabbit anti-mouse/human OLIG2 A647 (EPR2673)	Abcam	ab225100; RRID: AB_10861310
Mouse anti-H3K27ac (clone MAB1 0309)	Active Motif	39085; RRID: AB_2793305
Mouse anti-MITF	Active Motif	91201; RRID: AB_2793801
Goat Anti-TFEB Polyclonal	Abcam	ab2636; RRID:AB_303224
Rabbit Anti-TFEB (D2O7D) mAb	Cell Signaling	37785; RRID:AB_2799119
Rabbit Anti-TFE3	Abcam	ab93808; RRID:AB_10563130
Rabbit Anti-TFE3	MilliporeSigma	HPA023881; RRID:AB_1857931
Rabbit Anti-TFE3	Novus	NBP1-89976; RRID:AB_11011044
Rabbit anti-Iba1	FujiFilm	019-19741; RRID:AB_839504
Rat anti-Cd68	Bio-Rad	MCA1957; RRID:AB_322219
Donkey anti-rabbit 555	ThermoFisher	A32794; RRID:AB_2762834
Donkey anti-rat 650	ThermoFisher	SA5-10029; RRID:AB_2556609
Chemicals, peptides, and recombinant proteins		
5% Donkey serum	Millipore Sigma	D9663
4',6-Diamidino-2-phenylindole, diacetate (DAPI)	BioLegend	422801
Dynabeads Protein A	ThermoFisher	10002D
Dynabeads Protein G	ThermoFisher	10004D
SpeedBeads magnetic carboxylate modified particles	Cytiva	GE45152105050250
TRIzol LS Reagent	ThermoFisher	10296028
Formaldehyde, 37% by weight	ThermoFisher	F79-1
Dulbecco's PBS (DPBS) solution	ThermoFisher	MT21031CV
Disuccinimidyl glutarate (DSG)	ProteoChem	C1104
UltraPure DNase/RNase-free distilled water	ThermoFisher	10977023
Oligo d(T)25 Magnetic Beads	NEB	S1419S
SUPERase-In	Ambion	AM2696
Oligo dT primer	ThermoFisher	18418020
Agencourt RNA Clean XP Beads	BeckmanCoulter	A63987
10 X Blue Buffer	Enzymatics	P7050L
DNA polymerase I	Enzymatics	P7050L
Random primers	ThermoFisher	48190011
SuperScript III Reverse Transcriptase	ThermoFisher	18080044
5X first-strand buffer	ThermoFisher	18080044
Actinomycin D	Sigma	A1410
NEBNext High-Fidelity Q5 2X PCR Master Mix	NEB	M0544S
Proteinase inhibitor cocktail	Sigma-Aldrich	P8340
Glycine	MilliporeSigma	4810
1M Tris-HCl, pH 8.0	ThermoFisher	15568025

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<i>Continued</i>		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
0.5 M EDTA, pH 8.0	ThermoFisher	15575020
0.5 M EGTA, pH 8.0	ThermoFisher	405200081
1M MgCl ₂	ThermoFisher	AM9530G
Sucrose	ThermoFisher	S6500
Triton X-100	MilliporeSigma	T8787
1,4-Dithiothreitol (DTT)	ThermoFisher	BP172-5
Bovine serum albumin	MilliporeSigma	A3059
RNasin	Promega	N2115
LysoTracker Red	Invitrogen	DND-99
10% Sodium deoxycholate	ThermoFisher	40430018
20% N-lauroylsarcosine sodium salt solution	MilliporeSigma	L7414
8M Lithium chloride	MilliporeSigma	L7026
IGEPAL CA-630	MilliporeSigma	I8896
<i>Critical commercial assays</i>		
RNeasy Plus Micro kit	Qiagen	74034
Qubit dsDNA HS Kit	ThermoFisher	Q32851
Nextera DNA Library Prep Kit	Illumina	FC-121-1030
NEBNext Ultra II DNA library prep kit	NEB	E7645L
ChIP DNA Clean and Concentrator Kit	Zymo Research	D5205
Chromium Single Cell 3' Library & Gel Bead Kit v3	10X Genomics	PN-1000075
Chromium Single Cell 3' Library Construction Kit v3	10X Genomics	PN-100078
Chromium Chip B Single Cell Kit	10X Genomics	PN-1000153
Single Index Kit T Set A	10X Genomics	PN-1000213
Chromium Next GEM Single Cell 3' HT, Reagent Kits v3.1	10X Genomics	PN-1000348
Chromium Next GEM Chip M Single Cell Kit	10X Genomics	PN-1000349
Dual Index Kit TT Set A	10X Genomics	PN-1000215
NEBNext Ultra II Library Preparation Kit	NEB	E7645L
SuperScript III First-Strand Synthesis System	ThermoFisher	18080051
<i>Deposited data</i>		
ATAC-seq	This Paper	GEO: GSE278447
ChIP-seq	This Paper	GEO: GSE278448
RNA-seq	This Paper	GEO: GSE278449
scRNA-seq & snRNA-seq	This Paper	GEO: GSE278451
Bulk RNA-seq of mucopolipidosis type IV (reanalyzed)	Cougnoux et al. ⁷⁴	GEO: GSE112895
Bulk RNA-seq of MPS type II microglia (reanalyzed)	Ullman et al. ⁶⁸	GEO: GSE139002
Bulk RNA-seq of Clec7a ^{hi} /Clec7a ^{lo} APP/PS1 microglia (reanalyzed)	Krasemann et al. ⁷³	GEO: GSE101689
Tau/ApoE4 reactive microglia (TERM) (reanalyzed)	Gratuzze et al. ¹⁰²	GEO: GSE206368
ATAC-seq of APP/PS1 microglia	Schlachetzki et al. ¹⁰⁸	GEO: GSE278028
snRNA-seq of human cortical biopsies (NPH cohort, reanalyzed)	Gazestani et al. ¹⁰³	https://braincelldata.org/resource
snRNA-seq of human cortical postmortem cohort, reanalyzed)	Green et al. ¹⁰⁹	AD Knowledge Portal (Synapse): syn31512863

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Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Organisms/strains		
Mouse: C57BL/6J	Jackson Laboratory	Strain #:000664, RRID:IMSR_JAX:000664
Mouse: <i>Sgsh</i> ^{Δexon3-6} (germline)	Esko and Gordts labs (UCSD)	N/A
Oligonucleotides		
Non-targeting control ASO	Ionis Pharmaceuticals	N/A
<i>Mitf</i> ASO	Ionis Pharmaceuticals	N/A
<i>Tfeb</i> ASO	Ionis Pharmaceuticals	N/A
<i>Tfe3</i> ASO	Ionis Pharmaceuticals	N/A
Software and algorithms		
Python (v3.1)	Python Software Foundation	https://www.python.org
Bowtie2 (v2.3.5.1)	Langmead and Salzberg ¹²⁴	https://bowtie-bio.sourceforge.net/bowtie2/index.shtml
STAR (v2.5.3a)	Dobin et al. ¹²⁵	https://github.com/alexdobin/STAR
HOMER (v4.11)	Heinz et al. ⁸⁴	http://homer.ucsd.edu/homer/
Irreproducibility Discovery Rate (IDR)	Li et al. ⁸²	www.encodeproject.org/software/idr/
CARTMAN	Saisan and Glass ⁹³	https://github.com/psaisan/CARTMAN/tree/main
Kallisto (v0.44.0)	Bray et al. ¹²⁶	https://pachterlab.github.io/kallisto
R (v4.2.3)	N/A	https://cran.r-project.org/
R package: tximport	Soneson et al. ¹²⁷	https://bioconductor.org/packages/release/bioc/html/tximport.html
R package: DESeq2 (v1.36.0)	Love et al. ¹²⁸	https://bioconductor.org/packages/release/bioc/html/DESeq2.html
R package: pheatmap (v1.0.12)	Kolde ¹²⁹	https://CRAN.R-project.org/package=pheatmap
Metascape	N/A	http://metascape.org/gp/index.html#/main/step1
10x Genomics Cell Ranger (v7.2.0)	Zheng et al. ¹³⁰	https://www.10xgenomics.com/support/software/cell-ranger/latest
R package: Seurat (v4.4.0)	Butler et al. ¹³¹	https://CRAN.R-project.org/package=Seurat
R package: scCustomize (v2.1.2)	Marsh ¹³²	https://cran.r-project.org/package=scCustomize
R package: scDbfFinder	Germain et al. ¹³³	https://bioconductor.org/packages/release/bioc/html/scDbfFinder.htm
CellBender	Fleming et al. ¹³⁴	https://cellbender.readthedocs.io/en/latest/
scRNAseqquest	Li et al. ¹³⁵	https://github.com/interactivereport/scRNAseqquest
Harmony	Korsunsky et al. ¹³⁶	https://github.com/immunogenomics/harmony
NEBULA	He et al. ⁶⁷	https://github.com/lhe17/nebula
FlowJo (v10)	N/A	https://www.flowjo.com/
GraphPad Prism 10	N/A	https://www.graphpad.com/
Fiji-ImageJ (v1.54f)	Schindelin et al. ¹³⁷	https://imagej.net/ij/
Other		
BD Influx	BD Biosciences	Equipment
MoFlo Astrios	BeckmanCoulter	Equipment
PIXUL Multi-Sample Sonicator	Active Motif	Equipment
UCSD Cellular and Molecular Medicine Electron Microscopy Core (JEOL 1400-plus TEM)	JEOL	Equipment; RRID:SCR_022039

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Mice

All animal procedures were approved by the University of California San Diego Animal Care and Use Committee in accordance with the University of California San Diego research guidelines for the care and use of laboratory animals. The following mouse strains were used in this study: C57BL/6J (The Jackson Laboratory, Stock No. 000664) and germline *Sgsh*^{-/-59} (Dr. Jeffery Esko and Dr. Philip Gordts laboratories, University of California, San Diego). Mice were housed at the University of California San Diego animal facility on a 12-hour/12-hour light/dark cycle with free access to normal chow food and water.

METHOD DETAILS

Nuclei isolation for snRNA-seq

Forebrain was dissected from six-month-old WT, *Sgsh*^{+/-}, and *Sgsh*^{-/-} mice and olfactory bulbs removed. Dissected tissue was homogenized in lysis buffer (10% Triton X-100, 0.1 M DTT, 40 U/μL RNasin (Promega, N2115), protease inhibitor cocktail (Sigma-Aldrich, P8340), and 2% bovine serum albumin (BSA) (Sigma Aldrich, B6917) in PBS) using a Dounce homogenizer for 10 strokes. Samples were incubated for 5 minutes on ice and then centrifuged at 500xg for 5 minutes at 4°C in a tabletop centrifuge. Pellets were resuspended in sorting buffer (0.5 M EDTA, 40 U/μL RNasin, 2% BSA in PBS), transferred to FACS tubes, filtered through a 70 μm mesh strainer, and centrifuged at 800xg for 5 minutes at 4°C in a swinging bucket centrifuge. DAPI (1:1,000; BioLegend 422801) was used to label nuclei. Samples were strained through a 70 μm mesh strainer, washed once more, resuspended in sorting buffer, and kept on ice until nuclei sorting. Samples were sorted for DAPI⁺ singlets on a Beckman Coulter MoFlo Astrios EQ.

Single nucleus RNA-seq and analysis

Sorted purified nuclei were centrifuged at 800xg for 5 minutes at 4°C and then resuspended in PBS supplemented with 400 mg/mL BSA. Appropriate volumes of each sample were loaded onto a Chromium Controller (10x Genomics) to couple to beads targeting 20,000 cells per sample per the manufacturer's Single Cell 3' HT v3.1 (HT) protocol. Following capture, single nuclei sequencing libraries were prepared following the 10X Genomics HT protocol. Samples were sequenced on an Illumina NovaSeq X platform targeting 50,000 reads per cell.

Raw FASTQ files for each sample were processed through Cell Ranger v7.2.0 (10x Genomics) with standard parameters aligning to the mm10-2020-A reference transcriptome. Downstream analysis was performed with the single cell analysis suite 'scRNA-seq-quest'¹³⁵. To remove ambient RNA contamination, CellBender¹³⁴ was used with the command `scRMambient` using the 'expected_cells' values equivalent to the captured cell values from 10x and the 'droplets_included' values ~25-50% higher than the 10x captured cell values to ensure capturing empty droplets. CellBender-corrected h5 files were then ran through the scRNA-seq pipeline with the command `scAnalyzer`; keeping genes expressed in at least 20 nuclei, containing 100-5,500 genes, a maximum 18,000 UMIs, and a doublet score of less than 0.02. Normalization, dimensionality reduction, and integration were carried out by Seurat^{131,138,139} and Harmony (v1.2.0).¹³⁶ Cluster/cell type-specific differential gene expression analyses were carried out in clusters containing a total of >500 nuclei across genotypes using NEBULA (v1.5.3),⁶⁷ specifically the NEBULA-HL algorithm. Total astrocytes recovered did not exceed 500 and thus were not analyzed. Data visualization of data and figure generation was performed in CellxGene.¹⁴⁰

Electron microscopy and analysis

Samples were processed and prepared by the UCSD CMM Electron Microscopy Facility. For EM on cortex sections, mice were transcardially perfused with Ringer solution followed by 10-15 mL of Fixative Solution (2% paraformaldehyde / 2.5% glutaraldehyde in 0.15 M sodium cacodylate buffer). Brains were quickly removed, dissected into hemispheres, and post-fixed in Fixative Solution at room temperature for 2-4 hours then overnight at 4°C. They were then incubated in 1% OsO₄ in 0.1 M cacodylate buffer for 1 hour on ice and then stained with 2% uranyl acetate for 1 hour on ice. The samples were dehydrated in a graded series of ethanol washes (50%-100%) once, followed by a wash with 100% ethanol and 2 washes with acetone for 15 minutes each, and embedded with Durcupan. 70 nm sections were cut on a Leica UCT ultra-microtome and collected on 300 mesh copper grids. Sections were stained with 2% uranyl acetate for 5 minutes, and Sato lead stain for 1 minute. Samples were viewed using a JEOL 1400-plus TEM (JEOL, Peabody, MA). Transmission electron microscopy images were taken using a Gatan OneView 4kx4k camera (Gatan, Pleasanton, CA). Vacuole measurement was performed with the Fiji-ImageJ v1.54f software. For vacuole quantification, cytoplasmic/perinuclear vacuoles >1000 nm in diameter were counted in 8-21 microglia or non-microglial cells per mouse (Total counts: WT^(MG)=35, WT^(Non-MG)=43, *Sgsh*^{-/-}(MG)=39, *Sgsh*^{-/-}(Non-MG)=72 cells). Counts were averaged per mouse for each cell class, log1p-transformed, and analyzed in Prism 10.2.1 (GraphPad) using a two-way mixed-effects model with cell type paired within mouse and genotype compared between mice, followed by Šidák-adjusted planned comparisons (WT vs *Sgsh*^{-/-} within each cell type; MG vs Non-MG within each genotype).

For EM on sorted ex-vivo microglia, cells were pelleted and chemically fixed with freshly prepared fixative (2.5% glutaraldehyde, in 0.1 M cacodylate buffer pH 7.4). They were then post-fixed in 1% OsO₄ in 0.1 M cacodylate buffer for 1 hour on ice and then stained with 2% uranyl acetate for 1 hour on ice. The samples were dehydrated in a graded series of ethanol washes (50%-100%) once, followed by a wash with 100% ethanol and 2 washes with acetone for 15 minutes each, and embedded with Durcupan. 70 nm

sections were cut on a Leica UCT ultra-microtome and collected on 300 mesh copper grids. Sections were stained with 2% uranyl acetate for 5 minutes, and Sato lead stain for 1 minute. Samples were viewed using a JEOL 1400-plus TEM (JEOL, Peabody, MA). Transmission electron microscopy images were taken using a Gatan OneView 4kx4k camera (Gatan, Pleasanton, CA). Quantification was performed with Fiji-ImageJ v1.54f software. For microglia cell areas, 20 cells across 2–4 FOVs were traced and quantified in each replicate for each genotype (WT=60, *Sgsh*^{−/−}=60 cells). For vacuole quantification, cytoplasmic vacuoles greater than 1000 nm in diameter at their widest cross section were counted in 20–40 cells in each replicate for each genotype (WT=100, *Sgsh*^{−/−}=63 cells). Statistical analyses were performed using unpaired two-tailed t-tests in Prism 10 (GraphPad).

Immunofluorescence and analysis

WT and *Sgsh*^{−/−} mice were perfused and fixed with 4% paraformaldehyde in PBS. Samples were sectioned (Leica CM3050 S) at 16 μm and collected on Superfrost Plus slides. Sections were fixed with 4% PFA for 10 min, permeabilized in PBS-T (PBS with 0.3% Triton X-100), and blocked with 5% donkey serum (Sigma, D9663) for 2 hours at room temperature. Slides were incubated with primary antibodies 1:1000 rabbit anti-Iba1 (Fujifilm, 019-19741) and 1:300 rat anti-Cd68 (Bio-Rad, MCA1975) at 4°C overnight, then incubated with 1:500 secondary antibodies (donkey anti-rabbit 555, Invitrogen A32794; donkey anti-rat 650, Invitrogen SA5-10029) for 2 hours at room temperature, counter-stained with DAPI, and mounted with Prolong Gold antifade reagent. Images were acquired on a Leica SP8 Confocal microscope with a 63x objective and processed with ImageJ. 63x z-stack FOVs across cerebral cortices of two mice per genotype were analyzed. Raw images were max projected, binarized, and thresholded with Otsu's Method. Cd68⁺ staining areas within Iba1⁺ microglia were quantified. 85 individual cells were measured over 10–16 FOVs per group. 'Fill Holes' command was used to fill in cell somas before measuring microglia soma sizes. Microglia branching was quantified using NeuronJ¹⁴¹ measuring 54 individual cells over 6–11 FOVs per group. Statistical analyses were performed using unpaired two-tailed t-tests in Prism 10 (GraphPad).

Isolation of live microglia

Forebrain from mice were homogenized and microglia isolated as previously described.^{83,142} Briefly, forebrains were dissected and olfactory bulbs removed. Tissue was homogenized by gentle mechanical dissociation, filtered, washed, microglia enriched with a Percoll gradient, and washed again. All steps were maintained at 4°C with the exception of the Percoll spin. Cells were then incubated in staining buffer on ice with anti-CD16/32 blocking antibody (BioLegend 101319) for 15 minutes, then with anti-mouse anti-CD11b-APC (BioLegend 101212), anti-CD45-Alexa488 (BioLegend 103122), and anti-CX3CR1-PE (BioLegend 149006) for 20 minutes. Cells were then washed and filtered through a 40 μm cell strainer. Sorting was performed on a Sony MA900 or Beckman Coulter MoFlo Astrios EQ cell sorter. Microglia were defined as events that were DAPI negative, singlets, and CD11b⁺/CD45^{int}/CX3CR1⁺. Isolated microglia were then processed according to protocols for RNA-seq, ATAC-seq, LysoTracker or EM.

Single cell RNA-seq and analysis

CD45⁺ sorted cells from wildtype and *Sgsh*^{−/−} mice were spun down at 400xg for 5 minutes, supernatant carefully aspirated, and resuspended to 60 μL PBS supplemented with 400 μg/mL BSA. Appropriate volumes of each sample were loaded onto a Chromium Controller (10x Genomics) to target 10,000 cells per sample. Following capture, single cell sequencing libraries were generated using the Chromium Single Cell 3' Library Reagent Kit (v3 Chemistry, 10X Genomics) following manufacturer's instructions. Samples were sequenced on an Illumina NovaSeq platform targeting 50,000 reads per cell.

Raw FASTQ files for each sample were processed through Cell Ranger v7.2.0 with standard parameters aligning to the mm10-2020-A reference transcriptome. Lower than expected cell capture rates led to mean reads per cell increasing 2 to 3-fold per sample. Resulting filtered matrices were analyzed in R v4.2.3 using Seurat v4.4.0.¹³¹ QC metrics were added to the Seurat object using the package scCustomize (v2.1.2). Genes expressed in less than 20 droplets were discarded, and droplets meeting the following criteria were kept for downstream analysis: containing >400 and <9500 genes, UMIs >1500, mitochondrial genes <10%, and a complexity ($\log_{10}(\text{nFeature}) / \log_{10}(\text{nCount})$) >0.78 and <0.92. The scDbfFinder package was used to identify and remove putative doublets.¹³³ All samples were processed uniformly through one 10x Chromium Chip and library preparation and sequenced together on one Illumina flowcell. No significant batch effects were observed thus integration was not performed. Normalization and dimensionality reduction were carried out according to the standard Seurat v4 pipeline. Major cell types were determined by clustering and known marker gene analysis. The microglia cluster was isolated and subclustered and the package dittoSeq (v1.8.1) was used for comparing cluster distributions. Differential gene expression analysis between genotypes in the microglia cluster was carried out using NEBULA (v1.5.3),⁶⁷ specifically the NEBULA-HL algorithm.

Sorted microglia RNA sequencing

Total RNA was isolated from 100–300K sorted microglia with RNeasy Plus micro (Qiagen, 74034) or Direct-zol RNA MicroPrep (Zymo Research, R2052) kits and stored at −80°C until RNA-seq library preparation. RNA-seq libraries were prepared as previously described.^{83,84} Briefly, polyA mRNA was enriched by two successive rounds of incubation with Oligo d(T) Magnetic Beads (NEB, S1419S) and then fragmented/eluted by incubation at 94°C for 9 min. cDNA was then synthesized with Superscript III (first-strand synthesis; Life Technologies kit 18080-044) and then DNA Polymerase I (second-strand synthesis; Enzymatics P7050L) with dUTP. Resulting cDNA libraries were prepared for sequencing by blunting/end repair, A-tailing, and barcoded adapter ligation (NEXTflex, Bioo Scientific/PerkinElmer) as previously described in Heinz et al.⁸⁴ The resulting RNA-seq libraries were digested by

incubation with Uracil DNA Glycosylase for 30 min at 37°C. Libraries were PCR amplified for 11–14 cycles, size selected for 200–500 bp fragments by gel excision, quantified using a Qubit dsDNA HS Assay Kit (Thermo Fisher Scientific) and single-end sequenced on a HiSeq 4000 or NextSeq 500 (Illumina, San Diego, CA) according to the manufacturer's instructions.

Bulk RNA-, ATAC-, and ChIP-seq preprocessing and mapping

RNA-seq FASTQ files were mapped using STAR (v2.5.3a)¹²⁵ using the mm10 genome and default parameters. For ASO RNA-seq experiments, FASTQ files were pseudomapped using Kallisto (v0.44.0)¹²⁶ with `-single -l 200 -s 30 -rf-stranded -single-overhang -b 50` parameters. ATAC- and ChIP-seq data were mapped with Bowtie2 (v2.3.5.1),¹⁴³ with ATAC-seq reads trimmed to 30bp prior to mapping. After mapping, HOMER (v4.11)⁸⁴ was used to convert reads aligning to a single genomic location into “tag directories” for subsequent analysis using the command `makeTagDirectory`. HOMER was also used to generate sequencing-depth normalized bedGraph/bigWig files for visualization in the UCSC Genome Browser (<http://genome.ucsc.edu>). A summary of sequencing QC metrics can be found in Tables S3 and S4.

RNA-seq analysis

Raw gene expression counts from tag directories were quantified by HOMER's `analyzeRepeats` command with parameters “-condenseGenes -count exons -noadj”, and TPM counts with parameters “-count exons -tpm”. TPM counts were matched to raw counts using accession numbers. Small RNAs (<200 bp) and genes with TPM<3 in all conditions were removed from analysis. Differential gene expression was calculated with the HOMER command `'getDiffExpression.pl'` utilizing DESeq2¹²⁸ (v1.36.0) and DEGs were accessed at FC>2 and p.adj<0.05 unless otherwise indicated. DEGs attributable to trace contamination from non-microglial cell types (e.g., neutrophils) in FACS-sorted bulk microglial RNA-seq were removed using reference single cell RNA-seq data from WT and *Sgsh*^{-/-} mice. Genes were excluded if they were (1) highly enriched in non-microglial clusters (min.FC>1, p.adj<0.05) and (2) were not expressed in WT or *Sgsh*^{-/-} microglia clusters (<1.5–2.5% of total microglia). For ASO RNA-seq experiments, abundance files from Kallisto pseudoalignment were imported with `tximport`¹²⁷ (v1.24.0) and processed with DESeq2 (v1.36.0) controlling for batch (design = ~batch + treatment) and keeping genes where the sum of normalized counts in each treatment group was >50 counts. Resulting filtered matrices were further analyzed in R v4.2.3. Heatmaps were generated using the package `ph heatmap` (v.1.0.12). Pathway enrichment analysis was performed with Metascape using Gene Prioritization by Evidence Counting (GPEC) and GO Biological Process, Reactome, and KEGG pathway gene sets (v3.5).¹⁴⁴ A summary of RNA-seq sequencing QC metrics can be found in Table S3.

LysoTracker assay on live microglia and analysis

Live sorted microglia were pelleted at 400xg for 5 minutes at 4°C following sorting. Supernatant was removed and the pellets resuspended in 1 mL of RPMI media without Phenol Red (Gibco, 11835-030) prewarmed to 37°C. 100 µL of the microglia in media was removed to serve as the negative control. LysoTracker Red (Invitrogen, DND-99) was added to samples at 1:500 to a final concentration of 2 nM and incubated at 37°C for 30 minutes. Following LysoTracker incubation, samples were spun down, media removed, and cells were resuspended in fresh warm RPMI media without LysoTracker. Samples and non-LysoTracker negative controls were run through a Sony MA900 measuring fluorescence on the PE-Texas Red channel. Quantification of LysoTracker signal (mean fluorescent intensity) was performed with the FlowJo software (v10.4.1).

Isolation of brain nuclei for ATAC/ChIP-sequencing

Brain nuclei were isolated as previously described,^{23,80} with initial homogenization performed with either 1% formaldehyde in PBS for 10 minutes (histone ChIP-seq assays), 0.1% formaldehyde in PBS for 10 minutes (oligodendrocyte and neuron ATAC-seq), or 2 mM DSG (Proteochem) for 30 minutes followed by 1% formaldehyde in PBS for 10 minutes (transcription factor ChIP-seq). Nuclei were stained overnight at 4°C with PU.1-PE (microglia) (Cell Signaling 81886S), OLIG2-AF488 (oligodendrocytes) (Abcam 225099), and NEUN-AF488 (neurons) (Millipore MAB 377X). Nuclei were washed with 4 mL FACS buffer, passed through a 40 µm strainer, and stained with 0.5 µg/mL DAPI (BioLegend 422801). Nuclei for each cell type were sorted with a Beckman Coulter MoFlo Astrios EQ or BD Influx Cell Sorter and pelleted at 1600xg for 5 minutes at 4°C in FACS buffer. Nuclei pellets were snap frozen and stored at -80°C prior to library preparation.

ATAC-seq

ATAC-seq libraries were prepared as previously described^{23,81,145,146} with approximately 50,000 sorted microglia. Cells were lysed in 150 µL lysis buffer (10 mM Tris-HCl pH 7.5, 10 mM NaCl, 3 mM MgCl₂ and 0.1% IGEPAL CA-630 in water). Resulting nuclei were centrifuged at 500xg for 10 min. For oligodendrocytes and neurons, approximately 200,000 lightly crosslinked sorted nuclei were used. Pelleted nuclei were resuspended in 50 µL transposase reaction mix (47.5 µL ATAC lysis buffer¹⁴⁶ and 2.5 µL Tagment DNA TDE1 enzyme I (Illumina, 20034197)) and incubated at 37°C for 30 min. DNA was purified with Zymo ChIP DNA concentrator columns (Zymo Research, D5205), eluted with 11 µL of elution buffer, and amplified using NEBNext High-Fidelity 2x PCR Master Mix (New England BioLabs, M0541) with the Nextera primer Ad1 (1.25 µM) and a unique barcoding primer Ad2 (1.25 µM) for 8–12 cycles. Resulting libraries were size selected by gel excision to 155–250 bp, purified and single-end sequenced using a HiSeq 4000 (Illumina) according to the manufacturer's instructions.

ChIP-seq

Chromatin immunoprecipitation sequencing was performed as previously described.^{84,147} For H3K27ac ChIP-seq, ~500,000 fixed nuclei were quickly thawed on ice and resuspended in ice-cold LB3 buffer (10 mM Tris-HCl pH 7.5, 100 mM NaCl, 1 mM EDTA, 0.5 mM EGTA, 0.1% Na-Deoxycholate, 0.5% N-lauroylsarcosine) and 1x protease inhibitor cocktail. Nuclei / chromatin was sheared by sonication using the PIXUL Multi-Sample Sonicator (Active Motif) in low bind 96 well plates (Corning, #7007) following the manufacturer's instructions. Samples were sonicated with the following settings: Pulse (N): 50, PRF [kHz]: 1.00, Process time: 60 min, Burst Rate [Hz]: 20.00. Samples were pulse spun to 600xg at 4°C in the 96-well plate, transferred to low-bind tubes, and chromatin was recovered by retaining the supernatant following a spin at 21,000xg at 4°C for 10 minutes. The supernatant was then diluted 1.1-fold with ice-cold 10% Triton X-100. Two percent of the lysate was kept as ChIP input control. 10 μ L of Dynabead Protein A and 10 μ L of Protein G were added per sample, in addition to 2 μ g of a specific antibody for H3K27ac (Active Motif 39685). The samples were rotated overnight at 4°C and were washed as follows the next day: 3x with Wash Buffer I (20 mM Tris-HCl pH 7.5, 150 mM NaCl, 2 mM EDTA, 0.1% SDS, 1% Triton X-100) + 1x protease inhibitor cocktail, 3x with Wash Buffer III (10 mM Tris-HCl pH 7.5, 250 mM LiCl, 1% Triton X-100, 1 mM EDTA, 0.7% Sodium Deoxycholate) + 1x protease inhibitor cocktail, 2x with TET (0.2% Tween-20/TE) + 1/3x protease inhibitor cocktail, and 1x with IDTET (0.2% Tween-20, 10 mM Tris pH 8, 0.1 mM EDTA). Samples were finally resuspended in 25 μ L TT buffer (10 mM Tris pH 8 + 0.05% Tween 20) prior to on-bead library preparation. For MITF, TFEB, and TFE3 ChIP-seq, one million nuclei were thawed on ice and resuspended in ice-cold RLNR1 buffer (20mM Tris HCl pH 7.5, 150 mM NaCl, 1 mM EDTA, 0.5 mM EGTA, 0.4% sodium deoxycholate, 1% NP-40, 0.1% SDS, 0.5 mM DTT) + 1x protease inhibitor cocktail/PMSF. Nuclei / chromatin was sheared and recovered with the same method employed for H3K27ac. 10 μ L of Dynabead Protein A and 10 μ L of Protein G beads per sample were then coupled to either 2 μ g of MITF antibody (ActiveMotif, 91201), 2 μ g each of two TFEB antibodies (Abcam ab2636, Cell Signaling CST 37785 (D2O7D), or 2 μ g each of two TFE3 antibodies (Abcamab93808, Millipore Sigma HPA023881, or NBP1-89976). Bead/antibody complexes were added to each sample and rotated overnight at 4°C. The samples were washed with the following buffers: 2x RLNR1 + PIC/PMSF/DTT buffer, 3x LWB-RCNR1 buffer (10 mM Tris HCl pH 7.5, 1 mM EDTA, 0.7% sodium deoxycholate, 1% NP-40, 250 mM LiCl) + PIC/PMSF, 2x TET buffer, 1x IDTET buffer, and then resuspended in 25 μ L TT for on-bead library preparation.

Sequencing libraries for ChIP and input samples were prepared with NEBNext Ultra II DNA library prep kit (NEB, E7645) reagents according to the manufacturer's protocol on beads suspended in 25 μ L TT (10 mM Tris/HCl pH 7.5, 0.05% Tween-20), with reagent volumes reduced by half. DNA was eluted and crosslinks reversed by adding 4 μ L 10% SDS, 4.5 μ L 5 M NaCl, 3 μ L EDTA, 4 μ L EGTA, 1 μ L proteinase K (20 mg/ml), 16 μ L water, incubating for 1 h at 55°C, then 30 minutes to overnight at 65°C. DNA was purified using 2 μ L of SpeedBeads (Cytiva GE45152105050250) in a 20% PEG8000 / 1.5M NaCl solution to final concentration of 12% PEG, then eluted with 25 μ L TT. DNA contained in the eluate was then amplified for 12-14 cycles in 25 μ L PCR reactions using NEBNext High-Fidelity Q5 2X PCR Master Mix (NEB) and 0.5 mM each of primers Solexa 1GA and Solexa 1GB. Resulting libraries were size selected by gel excision to 200-500 bp, purified, and single end sequenced using an Illumina NovaSeq instrument.

IDR analysis of ATAC and ChIP peaks

ATAC- and ChIP-seq experiments were performed in replicates with corresponding input experiments. Peaks were called using HOMER findPeaks for each tag directory with relaxed peak finding parameters '-L 0 -C 0 -fdr 0.9'. ATAC-seq peaks were called with additional parameters '-minDist 200 -size 200'. Irreproducible Discovery Rate (IDR) (v2.0.4) was used to test for reproducibility between replicates⁸²; using only peaks with an IDR <0.05 for downstream analyses. For groups with more than two replicates, peak sets from all pairwise IDR comparisons were merged into a final set of peaks for further analysis. Encode blacklist regions were removed from final peak sets using the HOMER mergePeaks command with the '-prefix' parameter.

ATAC and ChIP analysis

To quantify differential regions of H3K27 acetylation between genotypes, IDR ATAC-seq peaks from WT and *Sgsh*^{-/-} microglia were merged with HOMER's mergePeaks command. Merged ATAC-seq peaks were then annotated with raw H3K27ac tag counts from each sample with HOMER's annotatePeaks using parameters -noadj and -size 2000. DESeq2 was used to identify the differential H3K27ac signal with the cutoffs of FC>2 and p.adj<0.05 using HOMER's getDiffExpression.pl. ATAC-seq peaks gaining H3K27ac were overlapped across time points using HOMER mergePeaks, which collapses overlapping peaks within and across samples into single merged loci. To identify motifs enriched in differential peak regions over background, HOMER's motif analysis (findMotifsGenome.pl) including known default motifs and *de novo* motifs was used. Background sequences were comprised of random GC-matched genomic sequences. For TFE3-bound promoter and enhancer analyses, promoters were defined as $\pm$ 500 bp from the TSS and enhancers as outside this region, both having $\geq$ 16 normalized merged tags for H3K27ac and $\geq$ 64 normalized tags for TFE3 IDR peaks.

Motif co-occurrence analysis

The motif co-occurrence analysis tool CARTMAN⁹³ was used to discover and differentially analyze motif co-occurrence patterns between two genomic peak sets. For the analysis presented in Figure 5F, *de novo* enriched motifs were identified for the 1272 TFE3 peaks exhibiting a >2-fold increase in H3K27ac in *Sgsh*^{-/-} microglia and the 3771 TFE3 peaks exhibiting significant (>16 normalized tags) but unchanged (<1.2 FC) H3K27ac using HOMER's findMotifsGenome.pl function.⁸⁴ Subsets of 5-8 of the most significantly enriched *de novo* motifs from each peak set were curated to form the target motif set. CARTMAN then annotated each peak with

motif presence/absence over a range of motif score thresholds and generated motif co-occurrence tables for all possible unique motif combinations at each threshold (2^n , where n is the number of motifs in the target set) and compared them for significant differences. CARTMAN provided normalized counts for each motif combination in both peak sets and assessed statistical significance using p-values derived via standard permutation testing.¹⁴⁸ Different subsets of the *de novo* motifs identified in TFE3 peaks with H3K27ac^{induced} and H3K27ac^{unchanged} were iteratively evaluated to estimate contributions of individual motifs to differential representation of specific combinations in the two target peak sets. As expected, the frequencies of specific motif co-occurrences increased as motif thresholds were reduced. However, most of the frequently observed differential motif combinations were consistently identified at similar levels of significance over a broad range of motif score thresholds. The analysis presented in Figure 5F is based on using the highest motif score threshold at which all peaks in both peak sets contain at least one of the five motifs (*i.e.*, the null set = 0).

Antisense Oligonucleotide (ASO) treatments

ASOs were diluted in PBS without Ca²⁺ and Mg²⁺ and administered via intracerebroventricular (ICV) bolus injection as previously described. Briefly, ASOs were delivered in a 10 μ L volume administered over 1 minute using the following coordinates: anteroposterior, +0.3 mm; mediolateral, +1.0 mm; and dorsoventral, -3.0 mm relative to bregma (right hemisphere). Single ASO treatment groups were administered with a 700 μ g dose and combination-ASO treatment groups were administered with a mixed solution of 500 μ g *Mitf*-ASO and 500 μ g *Tfe3*-ASO. All ASOs used were 5-10-5 MOE gapmers (designed and provided by Ionis Pharmaceuticals, Inc.). ASOs were designed to target the intron, exon or 3'-UTR of the mouse gene of interest. ASO sequences were as follows: Control ASO: CCTATAGGACTATCCAGGAA; mouse-specific TFE3-targeting ASO: ACACTTATTTTTTCTTCC; mouse-specific MITF-targeting ASO: TCTTCTTTTTTCTCTTGA; and mouse-specific TFEB-targeting ASO: CCTTCATTTTACCTTTACT.

snRNA-seq analysis of human microglia

We obtained the expression data of two large publicly available datasets that performed snRNA-seq on the frontal cortex of human cohorts. The first dataset is from Gazestani et al.,¹⁰³ which performed snRNA-seq on 52 surgical biopsies from patients undergoing ventriculoperitoneal shunt placement for the treatment of suspected normal pressure hydrocephalus (NPH). The second dataset is from Green et al.,¹⁰⁹ which studied single-nucleus RNA-sequencing profiles sampled from 437 older individual donors from the ROSMAP cohort. The UMAP representation and cell-type assignments were computed and extracted identically to the published results in the scanpy¹⁴⁹ framework. Aggregated enrichment scores of the bulk *ex vivo* microglia signature for the *Sgsh*^{-/-} genes (8 months; fold change >2, p.adj < 1e-10) for microglial states were calculated per nucleus using the "score_genes" function in scanpy, then averaged within each annotated cluster and z-scored across clusters (sklearn, StandardScaler), and then mapped back to nuclei in the original UMAP embeddings for each dataset.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses of sn/scRNA-seq, bulk RNA-seq and ChIP-seq data were performed as described above. All other data were analyzed using Prism 10 (GraphPad). Information regarding individual statistical tests is provided in each panel's figure legend.

Data visualization

UCSC genome browser (Kent et al.¹⁵⁰) was used to visualize ChIP-seq and ATAC-seq data, and Prism 10 (GraphPad) for bar and violin plots.